9. The figure shows three point charges kept at the vertices of triangle ABC. The net electric field, due to this system of charges, at the midpoint M of base BC will be :

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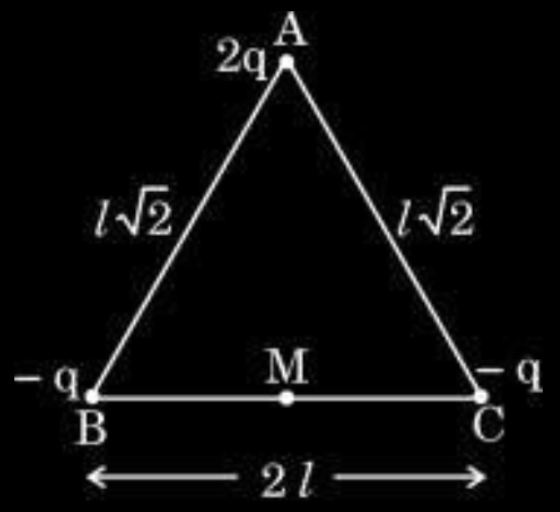


(A) $\frac{q}{4\pi\varepsilon_0 l^2}$ pointing along MA

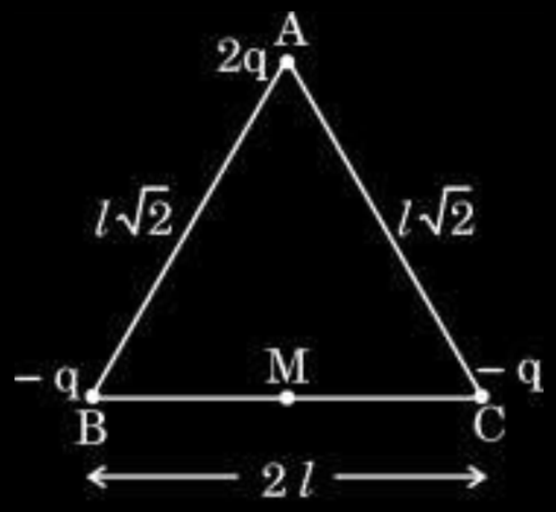
(B) $\frac{q}{\pi\varepsilon_0 l^2}$ pointing along AM

(C) $\frac{q}{2\pi\varepsilon_0 l^2}$ pointing along AM

(D) Zero

9. The figure shows three point charges kept at the vertices of triangle ABC. The net electric field, due to this system of charges, at the midpoint M of base BC will be :

CBSE 2026



(A) $\frac{q}{4\pi\varepsilon_0 l^2}$ pointing along MA (B) $\frac{q}{\pi\varepsilon_0 l^2}$ pointing along AM

(C) $\frac{q}{2\pi\varepsilon_0 l^2}$ pointing along AM (D) Zero

(C) $\frac{q}{2\pi\varepsilon_0 l^2}$ pointing along AM

25. Two point charges $q_1 = 2 \cdot 5 \times 10^{-7}$ C and $q_2 = -2 \cdot 5 \times 10^{-7}$ C are located at points (0, 0, – 15 cm) and (0, 0, 15 cm) respectively. Find :

(a) the electric dipole moment of the system, and

(b) the magnitude and direction of electric field at the origin (0, 0, 0). *3*

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25. Two point charges $q_1 = 2{\cdot}5 \times 10^{-7}$ C and $q_2 = -\,2{\cdot}5 \times 10^{-7}$ C are located at points (0, 0, – 15 cm) and (0, 0, 15 cm) respectively. Find :

(a) the electric dipole moment of the system, and

(b) the magnitude and direction of electric field at the origin (0, 0, 0). *3*

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(a) $p = q \times 2a$	
$= 2.5 \times 10^{-7} \times 30 \times 10^{-2}$	½
$= 75 \times 10^{-9}\,Cm$	½
(b)	
$E = \dfrac{1}{4\pi\epsilon_0}\dfrac{q}{r^2}$	½
$\lvert\vec{E}_1\rvert = \lvert\vec{E}_2\rvert = \dfrac{9 \times 10^9 \times 2.5 \times 10^{-7}}{(15 \times 10^{-2})^2} = 1 \times 10^5\,N/C$	½
$\vec{E} = \vec{E_1} + \vec{E_2} = 2\vec{E}_1 = 2\vec{E}_2 = 2 \times 10^5$ N/C	½
Direction: Along positive z-axis	½

6. Two point charges $-Q$ and Q are located at points (d, 0) and (0, d) respectively, in x-y plane. The electric field $\vec{E}$ at the origin will be :

(A) $\frac{1}{4\pi\varepsilon_0}\frac{\sqrt{2}\,Q}{d^2}(\hat{i}-\hat{j})$

(B) $\frac{1}{4\pi\varepsilon_0}\frac{\sqrt{2}\,Q}{d^2}(-\hat{i}-\hat{j})$

(C) $\frac{1}{4\pi\varepsilon_0}\frac{Q}{d^2}(\hat{i}-\hat{j})$

(D) $\frac{1}{4\pi\varepsilon_0}\frac{Q}{d^2}(-\hat{i}-\hat{j})$

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6. Two point charges $-Q$ and Q are located at points (d, 0) and (0, d) respectively, in x-y plane. The electric field $\vec{E}$ at the origin will be :

(A) $\frac{1}{4\pi\varepsilon_0}\frac{\sqrt{2}\,Q}{d^2}(\hat{i}-\hat{j})$

(B) $\frac{1}{4\pi\varepsilon_0}\frac{\sqrt{2}\,Q}{d^2}(-\hat{i}-\hat{j})$

(C) $\frac{1}{4\pi\varepsilon_0}\frac{Q}{d^2}(\hat{i}-\hat{j})$

(D) $\frac{1}{4\pi\varepsilon_0}\frac{Q}{d^2}(-\hat{i}-\hat{j})$

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(C) $\frac{1}{4\pi\varepsilon_0}\frac{Q}{d^2}(\hat{i}-j)$

7. An electron is moving with velocity $v_0 \hat{i}$. If a uniform electric field $\vec{E} = E_0 \hat{j}$ is set up in the region, the electron will :

(A) describe a circular path

(B) describe a helical path

(C) describe a parabolic path

(D) continue moving without any deviation

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7. An electron is moving with velocity $v_0 \hat{i}$. If a uniform electric field $\vec{E} = E_0 \hat{j}$ is set up in the region, the electron will :

(A) describe a circular path

(B) describe a helical path

(C) describe a parabolic path

(D) continue moving without any deviation

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(C) describes parabolic path

1. Two small identical metallic balls having charges q and – 2q are kept far at a separation r. They are brought in contact and then separated at distance $\frac{r}{2}$. Compared to the initial force F, they will now :

(A) attract with a force $\frac{F}{2}$

(B) repel with a force $\frac{F}{2}$

(C) repel with a force F

(D) attract with a force F

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1. Two small identical metallic balls having charges q and $-2q$ are kept far at a separation r. They are brought in contact and then separated at distance $\frac{r}{2}$. Compared to the initial force F, they will now :

(A) attract with a force $\frac{F}{2}$

(B) repel with a force $\frac{F}{2}$

(C) repel with a force F

(D) attract with a force F

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(B) repel with a force F/2

(b) (i) Two infinitely long straight wires having linear charge densities $-\lambda$ and 3λ are held vertically parallel to each other, distance r apart in free space. Find the nature and magnitude of the force/length exerted by one wire on the other.

(ii) A small hollow conducting sphere of radius r_1 is given a charge Q. It is surrounded by a concentric conducting spherical shell of inner radius r_2 and outer radius r_3, having charge $-3q$.

If a point charge 2q were kept at the centre, find :

(I) the electric flux through a concentric spherical Gaussian surface of radius x for (1) $x < r_1$, and (2) $r_1 < x < r_2$.

(II) electric field at a point distant x from the centre for (1) $x > r_3$, and (2) $r_1 < x < r_2$.

(III) surface charge density on the inner surface of (1) sphere, and (2) shell.

(i) Nature of force: Attractive	
$dq = \lambda dl$	½
Electric field on wire 2 due to wire 1, E = $\frac{-\lambda}{2\pi\varepsilon_0 r}$	
Force on wire 2 = qE	
$\frac{Force}{length} = \frac{F_1}{l} = \frac{-\lambda \times 3\lambda}{2\pi\varepsilon_0 r}$	½
$= \frac{-3\,\lambda^2}{2\pi\varepsilon_0 r}$	½
Similarly, $\frac{F_2}{l} = \frac{-3\lambda^2}{2\pi\varepsilon_0 r}$	
(ii) (I) (1) $x < r_1$	
$\phi = \frac{2q}{\varepsilon_0}$	½
(2) $r_1 < x < r_2$	
$\phi = \frac{(Q+2q)}{\varepsilon_0}$	½
(II) (1) $x > r_3$	½
$E = \frac{k\,(Q-q)}{x^2}$	
(2) $r_1 < x < r_2$	½
$E = \frac{k\,(Q+2q)}{x^2}$	½
(III) (1) $\sigma = \frac{-2q}{4\pi r_1^2}$	
	½
(2) $\sigma = \frac{-(Q+2q)}{4\pi r_2^2}$	

4. An electric dipole $\vec{P} = 2qa\,\hat{i}$ is placed in x-y plane, centred at the origin. Let E_1 and E_2 be the magnitudes of electric field at a point distant $r\,(>> a)$ from its centre, along its axis and on its equatorial plane, respectively. Then the value of $\left(\dfrac{E_1}{E_2}\right)$ is :

(A) $\dfrac{1}{4}$ (B) $\dfrac{1}{2}$

(C) 2 (D) 4

CBSE 2026

4. An electric dipole $\vec{P} = 2qa\,\hat{i}$ is placed in x-y plane, centred at the origin. Let E_1 and E_2 be the magnitudes of electric field at a point distant $r\ (>> a)$ from its centre, along its axis and on its equatorial plane, respectively. Then the value of $\left(\frac{E_1}{E_2}\right)$ is :

(A) $\frac{1}{4}$ (B) $\frac{1}{2}$

(C) 2 (D) 4

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(C) 2

1. A particle of mass m and charge q starts from rest and moves in an electric field $\vec{E} = E_0 \hat{i}$. After travelling a distance x in the field along x-axis, the kinetic energy of the particle will be :

(A) $q E_0 x^2$

(B) $q E_0 x$

(C) $q^2 E_0 x$

(D) $q^2 E_0^2 x$

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1. A particle of mass m and charge q starts from rest and moves in an electric field $\vec{E} = E_0 \hat{i}$. After travelling a distance x in the field along x-axis, the kinetic energy of the particle will be :

(A) $q E_0 x^2$

(B) $q E_0 x$

(C) $q^2 E_0 x$

(D) $q^2 E_0^2 x$

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(B) $qE_0 x$

1. How many protons are needed to make 1 nC charge ? 1

(A) $1.60 \times 10^{+28}$ (B) $1.60 \times 10^{+10}$

(C) 3.1×10^{9} (D) 6.25×10^{9} CBSE 2026

1. How many protons are needed to make 1 nC charge ? **1**

(A) $1.60 \times 10^{+28}$

(B) $1.60 \times 10^{+10}$

(C) 3.1×10^{9}

(D) 6.25×10^{9}

CBSE 2026

(D) 6.25×10^{9}

31. (a) Deduce the relation between the electric field ($\vec{E}$) at a point due to a charge Q and

(i) the force $\vec{F}$ on a charge q placed at that point.

(ii) the potential at that point due to charge Q.

(b) Three point charges, q each, are placed at the three vertices of an equilateral triangle of side 'a'. Obtain an expression for the net force on one charge due to the other two charges. **5**

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(i) (I) Electric field produced by the charge Q at a point r is given by:	
$\vec{E}(r)=\frac{1}{4\pi\varepsilon_o}\frac{Q}{r^2}\hat{r}=\frac{1}{4\pi\varepsilon_o}\frac{Q}{r^3}\vec{r}$	½
$\vec{F}(r)=\frac{1}{4\pi\varepsilon_o}\frac{Qq}{r^2}\hat{r}$	
$\vec{F}(r)=q\vec{E}(r)$	½
(II) Consider two closely spaced equipotential surfaces A and B, with potential values V and V+ δ V. Let δl be the perpendicular distance between equipotential surfaces A and B.	
The small amount of work done in moving a unit positive charge from the surface B to A along perpendicular distance δl,	½
$\delta W=\lvert\vec{E}\rvert\delta l=V-(V+\delta V)=-\delta V$	½
$\lvert\vec{E}\rvert=\frac{-\delta V}{\delta l}$	½
Since δV is negative,	
$\therefore\ \lvert\vec{E}\rvert=\frac{-\delta V}{\delta l}=+\frac{\lvert\delta V\rvert}{\delta l}$	½

(ii) Forces due to two charges on one charge at the vertices of equilateral triangle will be repulsive, The angle between the forces will be 60°, So, net force	½
$F_{net} = \sqrt{F_1^2 + F_2^2 + 2F_1F_2\cos 60^\circ}$	½
$F_1 = F_2 = F$ (as charges are same)	
$F_{net} = \sqrt{2F^2 + 2F^2\cos 60^\circ}$ $= \sqrt{3F^2} = \sqrt{3}F$	½
$F_{net} = \sqrt{3}\left(\frac{1}{4\pi\varepsilon_0}\right)\frac{q^2}{a^2}$	½

22. (a) Using Gauss's law, deduce an experession for electric field at a point due to a uniformly charged infinite plane thin sheet. **3**

(b) Two large thin plane sheets, each having surface charge density σ, are held close and parallel to each other in air. What is the net electric field at a point (i) inside and (ii) outside, the sheets ?

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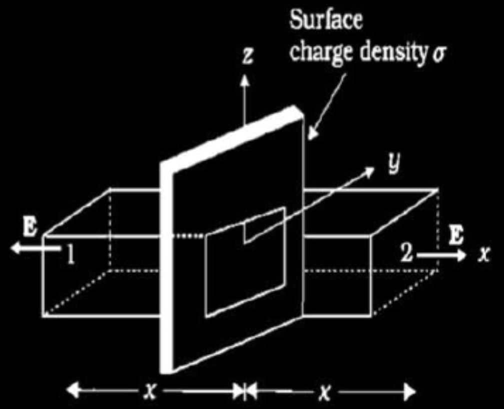
(a)



½

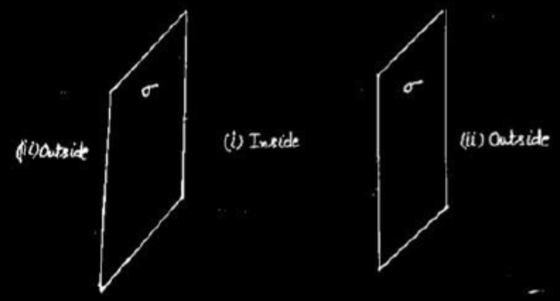
As seen from the figure, only the two faces 1 and 2 will contribute to the flux. Therefore, flux $\vec{E}.\Delta\vec{S}$ through both the surfaces are equal and add up. ½

Therefore, net flux through the Gaussian surface is 2 EA. The charge enclosed by the close surface is σ A.

$$2EA = \frac{\sigma A}{\varepsilon_0}$$ ½

$$\vec{E} = \frac{\sigma}{2\varepsilon_0}\hat{n}$$ ½

(b)

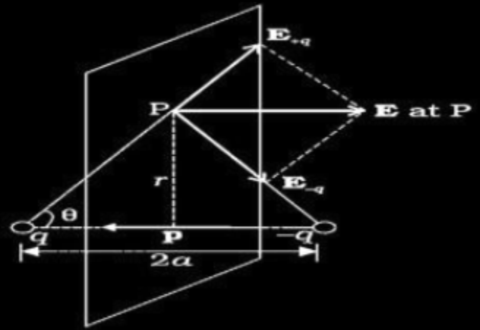


(i) $E_{in} = 0$ ½

(ii) $E_{out} = \frac{\sigma}{\varepsilon_0}$ ½

31. (a) An electric dipole consists of two point charges q and –q separated by a distance 2a. Derive an expression for the electric field $\vec{E}$ due to this dipole at a point distant r from the centre of the dipole on the equatorial plane. Write the expression for the electric field at a far off point, i.e. r >> a.

(b) A dipole is placed in x-y plane such that charges q and –q are located at x = a and x = b respectively. There exists an electric field $\vec{E} = 2\hat{i}\ \frac{N}{C}$ in the region. Calculate the force $\vec{F}$ and torque $\vec{\tau}$ experienced by the dipole. CBSE 2026 **5**



½

The magnitude of the electric fields due to the two charges + q and – q are given by

$$E_{+q} = \frac{q}{4\pi\varepsilon_0}\frac{1}{\left(r^2+a^2\right)}$$ ½

$$E_{-q} = \frac{q}{4\pi\varepsilon_0}\frac{1}{\left(r^2+a^2\right)}$$

The components normal to the dipole axis cancel away. The components along the dipole axis add up. The Total electric field is opposite to $\vec{p}$.

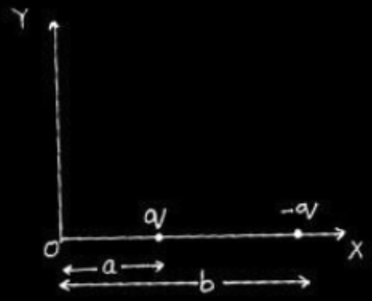
$$\vec{E} = -\left(E_{+q}+E_{-q}\right)\cos\theta\left(\hat{p}\right)$$ 1

$$\vec{E} = -\frac{1}{4\pi\varepsilon_0}\frac{2qa}{\left(r^2+a^2\right)^{3/2}}\hat{p}$$ ½

At large distance r ≫a

$$\vec{E} = \frac{-2qa}{4\pi\varepsilon_0 r^3}\hat{p}$$ ½

(b)



$$\because \vec{F} = \vec{F}_{+q} + \vec{F}_{-q}$$

Net force $= \left[+q.2\hat{i} - q.2\hat{i}\right]$ ½

$= 0\ N$ ½

Torque $\vec{\tau} = \vec{p} \times \vec{E}$

$\vec{\tau} = p\left(-\hat{i}\right) \times 2\hat{i}$ ½

$\tau = 0$ ½

Alternatively:

$\tau = pE\sin\theta$

Angle between $\vec{p}$ and $\vec{E}$ is π

$\tau = 0$

1. The charge on a body is 8×10^{-12} C. It means that the body has :

(a) lost 8×10^{-12} electrons

(b) gained 4×10^{10} electrons

(c) gained 2×10^{8} electrons

(d) lost 5×10^{7} electrons

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(d) Lost 5×10^{7} electrons

31. (a) (i) What is an electric dipole ? Derive an expression for the torque acting on an electric dipole in a uniform electric field.

(ii) An electric dipole with dipole moment 6×10^{-9} C-m is aligned at an angle of $30°$ with the direction of a uniform electric field of magnitude 4×10^{4} NC^{-1}. Calculate magnitude of the torque acting on the electric dipole. *5*

(i) Electric dipole-definition	1
Derivation of torque	2
(ii) Calculation of torque	2

(i) Electric Dipole:- An arrangement of two equal and opposite charges separated by a small distance. 1

Consider an electric dipole of charges $\pm q$ and length 2ℓ kept in a uniform electric field E. The dipole makes an angle θ with the uniform electric field. Forces on two charges due to uniform electric field, qE and $-qE$ are in opposite directions. The distance between the line of action of two force is $2l\sin\theta$. Torque due to the couple ½

$\tau =$ force $\times$ perpendicular distance between the two force ½

$$\tau = qE(2l\sin\theta) \quad ½$$

$$\tau = q(2l).E\sin\theta = pE\sin\theta \quad ½$$

(ii)

$$\tau = pE\sin\theta \quad ½$$

$$\tau = 6\times10^{-9}\times4\times10^{4}\times\sin 30^{o} \quad ½$$

$$\tau = 1.2\times10^{-4}\,\text{Nm} \quad 1$$

(b) (i) State Gauss's law in electrostatics. Using it, derive an expression for the electric field due to a uniformly charged thin spherical shell of radius R at a point (i) outside, and (ii) inside the shell.

(ii) A point charge of 4 μC is at the centre of a cubic Gaussian surface, 1·0 m on edge. Find the electric flux through one of the faces of the Gaussian surface. *5*

(b)

(i) Statement of Gauss's law in electrostatics.	1
Derivation of electric field due to a uniformly charged thin spherical shell	
(i) Outside the shell	1
(ii) Inside the shell	1
(ii) Calculation of electric flux.	2

(i) Gauss's law in electrostatics states that net electric flux linked with a closed surface equals $\frac{1}{\varepsilon_o}$ times the charge enclosed by the surface.

Consider a conducting spherical shell of radius R, charge given to it is q.

(i) For point P outside the shell construct a concentric sphere(Gaussian surface) of radius r (r>R)

Electric flux linked with the Gaussian sphere $\phi_E = E4\pi r^2$ ___________(i) 1

Charge enclosed by the Gaussian spherical surface = q

Gauss's Law

$$\phi_E = \frac{q}{\varepsilon_o}$$

½

$$E4\pi r^2 = \frac{q}{\varepsilon_o}$$

$$E = \frac{q}{4\pi\varepsilon_o r^2}$$

½

(ii) For a point inside the shell

Charge enclosed

by the Gaussian spherical surface of radius r' (r'<R)=0 ½

$$\therefore E 4\pi r'^2 = \frac{q_{in}}{\varepsilon_o} = 0$$

½

$$E = 0$$

(ii) Net electric flux linked with the cube

$$\phi'_{Enet} = \frac{q}{\varepsilon_o}$$ ½

Electric flux linked with one face

$$\phi'_E = \frac{\phi_{Enet}}{\varepsilon_o} = \frac{q}{6\varepsilon_o}$$ ½

$$\phi'_E = \frac{4\times10^{-6}}{6\times8.854\times10^{-12}}$$

$$= 7.5\times10^4\ Nm^2C^{-1}$$ 1

A small sphere S_1 with charge –8q is 1·6 m away from another identical sphere S_2 with charge +2q. The two spheres are brought in contact with each other and then separated by a distance 1·6 m. Initially the force between the two spheres was $8 \cdot 1 \times 10^{-4}$ N.

Based on the above facts, answer the following questions :

(i) Which sphere will transfer the electrons to the other sphere after they were brought in contact ? *1*

(ii) How does the net electric field at the midpoint on the line joining the two spheres change after contact ? *1*

(iii) What was initial charge on spheres S_1 and S_2 ? *2*

OR

(iii) What is the charge on spheres S_1 and S_2 after contact ? *2*

(i) Identification of sphere	1
(ii) Effect on net electric field after contact	1
(iii) Initial charge on S_1 and S_2	1+1
OR	
Charges on S_1 and S_2 after contact	1+1

(i) Electrons will be transferred from sphere S_1 to sphere S_2 when brought in contact. 1

(ii) net electric field changes from non zero to zero at the midpoint of line joining the centres of the two spheres. 1

(iii) $$\frac{1}{4\pi\varepsilon_0}\frac{q_1q_2}{r^2} = F$$ ½

$$9\times10^9 \times \frac{(8q)(2q)}{1.6\times1.6} = 8.1\times10^{-4}\,N$$

$$16q^2 = \frac{8.1\times1.6\times1.6\times10^{-4}}{9\times10^9}$$ ½

$$q^2 = \frac{8.1\times1.6\times1.6\times10^{-4}}{9\times10^9\times16}$$

$$q^2 = 9\times16\times10^{-16}$$

$$q = 12\times10^{-8}\,C$$ ½

Charge on S_1 = $-8q = -96\times10^{-8}$ C ½

Charge on S_2 = $2q = 24\times10^{-8}$ C ½

OR

(iii) Charges on S_1 and S_2 after were brought in contact

$-3q$ and $-3q$ 2

4. A particle of mass m and charge –q is moving with a uniform speed v in a circle of radius r, with another charge q at the centre of the circle. The value of r is :

(a) $\dfrac{1}{4\pi\varepsilon_0 m}\left(\dfrac{q}{v}\right)$

(b) $\dfrac{1}{4\pi\varepsilon_0 m}\left(\dfrac{q}{v}\right)^2$

(c) $\dfrac{m}{4\pi\varepsilon_0}\left(\dfrac{q}{v}\right)$

(d) $\dfrac{m}{4\pi\varepsilon_0}\left(\dfrac{q}{v}\right)^2$

4. A particle of mass m and charge –q is moving with a uniform speed v in a circle of radius r, with another charge q at the centre of the circle. The value of r is :

(a) $\dfrac{1}{4\pi \varepsilon_0 m}\left(\dfrac{q}{v}\right)$

(b) $\dfrac{1}{4\pi \varepsilon_0 m}\left(\dfrac{q}{v}\right)^2$

(c) $\dfrac{m}{4\pi \varepsilon_0}\left(\dfrac{q}{v}\right)$

(d) $\dfrac{m}{4\pi \varepsilon_0}\left(\dfrac{q}{v}\right)^2$

(b) $\dfrac{1}{4\pi\varepsilon_0 m}\cdot\left(\dfrac{q}{v}\right)^2$

6. A point charge q is kept at a distance r from an infinitely long straight wire with charge density λ. The magnitude of the electrostatic force experienced by charge q is :

(a) Zero

(b) $\dfrac{q\lambda}{2\pi\,\varepsilon_0\,r}$

(c) $\dfrac{q\lambda}{4\pi\,\varepsilon_0\,r}$

(d) $\dfrac{q\lambda}{\varepsilon_0\,r}$

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(a) Zero

(b) $\dfrac{q\lambda}{2\pi\,\varepsilon_0\, r}$

(c) $\dfrac{q\lambda}{4\pi\,\varepsilon_0\, r}$

(d) $\dfrac{q\lambda}{\varepsilon_0\, r}$

(b) $\dfrac{q\lambda}{2\pi \in_0 r}$

1. Two charges q_1 and q_2 are placed at the centres of two spherical conducting shells of radius r_1 and r_2 respectively. The shells are arranged such that their centres are d $[>(r_1+r_2)]$ distance apart. The force on q_2 due to q_1 is :

(a) $\dfrac{1}{4\pi\varepsilon_0}\dfrac{q_1 q_2}{d^2}$

(b) $\dfrac{1}{4\pi\varepsilon_0}\dfrac{q_1 q_2}{(d-r_1)^2}$

(c) Zero

(d) $\dfrac{1}{4\pi\varepsilon_0}\dfrac{q_1 q_2}{[d-(r_1+r_2)]^2}$

1. Two charges q_1 and q_2 are placed at the centres of two spherical conducting shells of radius r_1 and r_2 respectively. The shells are arranged such that their centres are d [> $(r_1 + r_2)$] distance apart. The force on q_2 due to q_1 is :

(a) $\frac{1}{4\pi\varepsilon_0}\frac{q_1q_2}{d^2}$

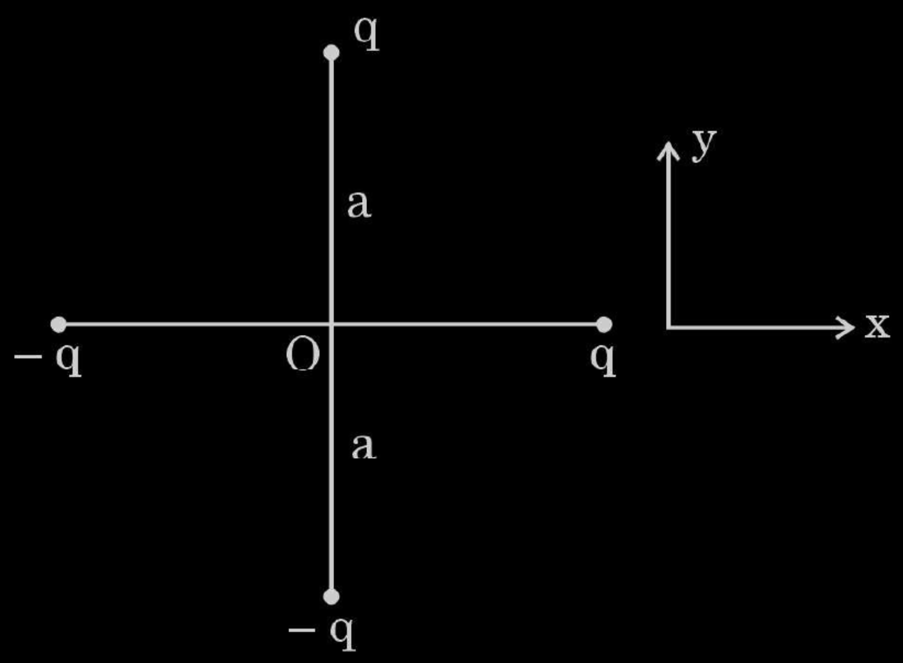
(b) $\frac{1}{4\pi\varepsilon_0}\frac{q_1q_2}{(d-r_1)^2}$

(c) Zero

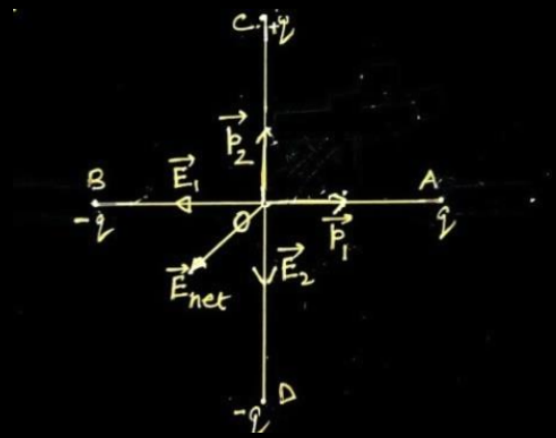
(d) $\frac{1}{4\pi\varepsilon_0}\frac{q_1q_2}{[d-(r_1+r_2)]^2}$

(c) zero

19. Two identical dipoles are arranged in x-y plane as shown in the figure. Find the magnitude and the direction of net electric field at the origin O. *2*



Finding the magnitude of electric field	1 ½
Finding direction of net Electric field	½



Dipole moment due to dipole BA is $\overrightarrow{p_1}$ & dipole moment due to dipole DC is $\overrightarrow{p_2}$.

Electric field $\overrightarrow{E_1}$ due to $\overrightarrow{p_1}$ is along OB.

Electric field $\overrightarrow{E_2}$ due to $\overrightarrow{p_2}$ is along OD. ½

Magnitude of resultant Electric field

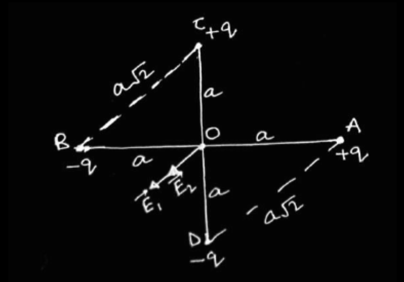
$\left|\overrightarrow{E_{net}}\right| = 2\sqrt{2}E$ ½

Since $\left|\overrightarrow{E_1}\right| = \left|\overrightarrow{E_2}\right| = E = \dfrac{1}{4\pi\epsilon_0}.\dfrac{q}{a^2}$

$\vec{E}_{net} = \dfrac{1}{4\pi\epsilon_0}.\dfrac{2\sqrt{2}\ q}{a^2}$ ½

Direction of $\overrightarrow{E_{net}}$ is 225° to x-axis. ½

Alternatively:



Considering CB as dipole, electric field at O

$$E_1 = \frac{2kq\times(a/\sqrt{2})}{\left[\left(\frac{a}{\sqrt{2}}\right)^2+\left(\frac{a}{\sqrt{2}}\right)^2\right]^{3/2}}$$

$$E_1 = \frac{\sqrt{2}kq\times a}{a^3\left(\frac{1}{2}+\frac{1}{2}\right)^{3/2}}$$

$$E_1 = \frac{\sqrt{2}kq}{a^2}$$ ½

Similarly considering AD as another dipole, electric field at O

$$E_2 = \frac{2kq\times(a/\sqrt{2})}{\left[\left(\frac{a}{\sqrt{2}}\right)^2+\left(\frac{a}{\sqrt{2}}\right)^2\right]^{3/2}}$$

$$E_2 = \frac{\sqrt{2}kq\times a}{a^3\left(\frac{1}{2}+\frac{1}{2}\right)^{3/2}}$$

$$E_2 = \frac{\sqrt{2}kq}{a^2}$$ ½

$$E_{net} = E_1 + E_2$$

$$= \frac{\sqrt{2}kq}{a^2}+\frac{\sqrt{2}kq}{a^2}$$

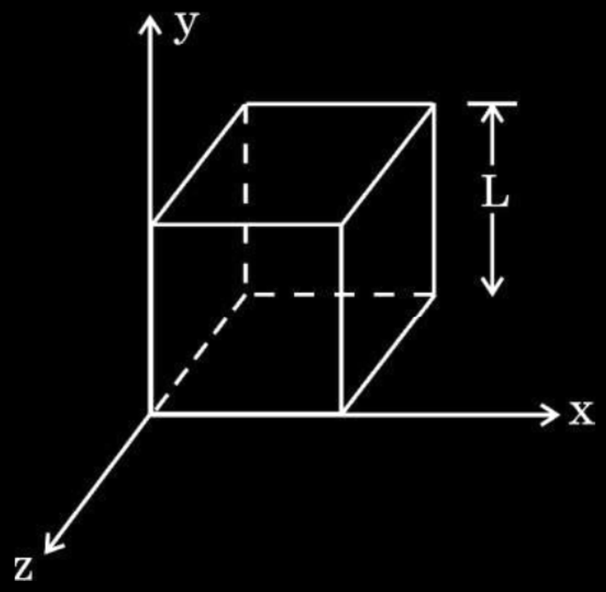
$$= \frac{2\sqrt{2}kq}{a^2}$$ ½

Direction of $\overrightarrow{E_{net}}$ is 225° to x-axis. ½

32. (a) (i) Define electric flux and write its SI unit.

(ii) Use Gauss' law to obtain the expression for the electric field due to a uniformly charged infinite plane sheet.

(iii) A cube of side L is kept in space, as shown in the figure. An electric field $\vec{E} = (Ax + B)\ \hat{i}\ \frac{N}{C}$ exists in the region. Find the net charge enclosed by the cube. 5



(a)

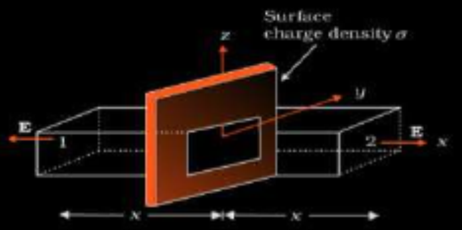
(i) Definition & SI Unit of Electric Flux	½+ ½
(ii) Deriving the expression for electric field due to a uniformly charged infinite plane sheet.	2
(iii) Net charge enclosed by the cube	2

(i) $\phi = \vec{E}.\vec{A}$ ½

Alternatively: Electric flux is the number of electric field lines passing through an area normally.

S.I. unit of electric flux Nm^2/C or V-m. ½

(ii)



½

From Gauss's law:- $\phi = \oint \vec{E} \,.\, \vec{dA} = \frac{q}{\varepsilon_o}$ ½

$2EA = \frac{\sigma A}{\varepsilon_o}$ ½

$E = \frac{\sigma}{2\varepsilon_o}$ ½

Alternatively: If the shape of the Gaussian surface is taken cylindrical, full credit to be given.

(iii)

$\phi_L = Eds\cos 180^o = -Eds$ ½

$= -BL^2$

$\phi_R = Eds\cos 0^0 = Eds$ ½

$= (AL + B)L^2 = AL^3 + BL^2$

Net flux $= \phi_L + \phi_R$ ½

$= (AL^3 + BL^2) - BL^2$

Net flux $= AL^3 = \frac{q}{\epsilon_o}$ ½

$\Rightarrow q = AL^3 \epsilon_o$

4. An infinitely long uniformly charged wire produces an electric field of 18×10^4 NC^{-1} at a distance of 1·0 cm. The linear charge density on the wire is :

(a) $1{\cdot}12 \times 10^{-14}$ Cm^{-1}

(b) $3{\cdot}08 \times 10^{-15}$ Cm^{-1}

(c) $1{\cdot}0 \times 10^{-9}$ Cm^{-1}

(d) $1{\cdot}0 \times 10^{-7}$ Cm^{-1}

4. An infinitely long uniformly charged wire produces an electric field of 18×10^4 NC^{-1} at a distance of 1·0 cm. The linear charge density on the wire is :

(a) $1{\cdot}12 \times 10^{-14}$ Cm^{-1} (b) $3{\cdot}08 \times 10^{-15}$ Cm^{-1}

(c) $1{\cdot}0 \times 10^{-9}$ Cm^{-1} (d) $1{\cdot}0 \times 10^{-7}$ Cm^{-1}

(d) 1.0×10^{-7} Cm^{-1}

1. An electron experiences a force $(1{\cdot}6 \times 10^{-16}\ \text{N})\ \hat{i}$ in an electric field $\vec{E}$. The electric field $\vec{E}$ is :

(a) $(1{\cdot}0 \times 10^{3}\ \frac{\text{N}}{\text{C}})\ \hat{i}$

(b) $-(1{\cdot}0 \times 10^{3}\ \frac{\text{N}}{\text{C}})\ \hat{i}$

(c) $(1{\cdot}0 \times 10^{-3}\ \frac{\text{N}}{\text{C}})\ \hat{i}$

(d) $-(1{\cdot}0 \times 10^{-3}\ \frac{\text{N}}{\text{C}})\ \hat{i}$

1. An electron experiences a force $(1 \cdot 6 \times 10^{-16}\ \mathrm{N})\ \hat{\mathrm{i}}$ in an electric field $\vec{\mathrm{E}}$. The electric field $\vec{\mathrm{E}}$ is :

(a) $(1 \cdot 0 \times 10^{3}\ \frac{\mathrm{N}}{\mathrm{C}})\ \hat{\mathrm{i}}$

(b) $-(1 \cdot 0 \times 10^{3}\ \frac{\mathrm{N}}{\mathrm{C}})\ \hat{\mathrm{i}}$

(c) $(1 \cdot 0 \times 10^{-3}\ \frac{\mathrm{N}}{\mathrm{C}})\ \hat{\mathrm{i}}$

(d) $-(1 \cdot 0 \times 10^{-3}\ \frac{\mathrm{N}}{\mathrm{C}})\ \hat{\mathrm{i}}$

(b) $-(1.0 \times 10^{3}\ \frac{N}{C})\hat{i}$

9. A point charge situated at a distance 'r' from a short electric dipole on its axis, experiences a force $\vec{F}$. If the distance of the charge is '2r', the force on the charge will be :

(a) $\frac{\vec{F}}{16}$

(b) $\frac{\vec{F}}{8}$

(c) $\frac{\vec{F}}{4}$

(d) $\frac{\vec{F}}{2}$

9. A point charge situated at a distance 'r' from a short electric dipole on its axis, experiences a force $\vec{F}$. If the distance of the charge is '2r', the force on the charge will be :

(a) $\dfrac{\vec{F}}{16}$

(b) $\dfrac{\vec{F}}{8}$

(c) $\dfrac{\vec{F}}{4}$

(d) $\dfrac{\vec{F}}{2}$

(b) $\dfrac{\vec{F}}{8}$

32. (a) (i) State Coulomb's law in electrostatics and write it in vector form, for two charges.

(ii) 'Gauss's law is based on the inverse-square dependence on distance contained in the Coulomb's law.' Explain.

(iii) Two charges A (charge q) and B (charge 2q) are located at points (0, 0) and (a, a) respectively. Let $\hat{i}$ and $\hat{j}$ be the unit vectors along x-axis and y-axis respectively. Find the force exerted by A on B, in terms of $\hat{i}$ and $\hat{j}$. *5*

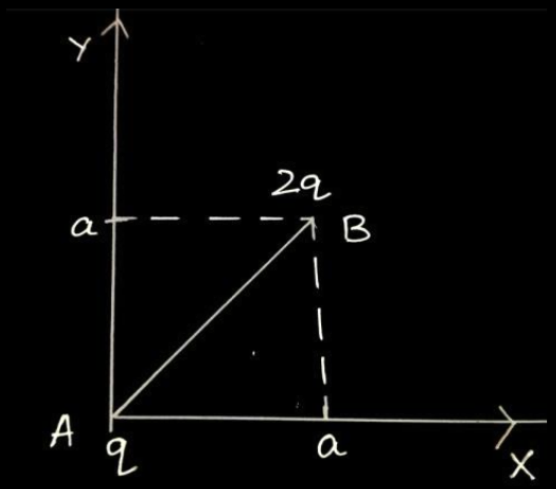
i) Statement of coulomb's law and vector form	1+1
ii) Explanation of Gauss's law based on coulomb's law	1
iii) Force exerted by charge A on charge B	2

i) Force between two point charges varies inversely with the square of distance between the charges and is directly proportional to the product of magnitude of the two charges and acts along the line joining the two charges.

$$\overrightarrow{F_{12}} = \frac{1}{4\pi \epsilon_o} \frac{q_1 q_2}{r_{12}^{\ 2}} \widehat{r_{12}}$$

ii) In derivation of Gauss's law, flux is calculated using Coulomb's law and surface area. Here coulomb's law involves $\frac{1}{r^2}$ factor and surface area involves r^2 factor. When product is taken, the two factors cancel out and flux becomes independent of r.

iii)



$$\vec{r} = \overrightarrow{AB} = a\hat{i} + a j$$

1 $\quad$ r = $\left|\overrightarrow{AB}\right| = \sqrt{a^2 + a^2} = \sqrt{2}a$

1 $\quad$ $\vec{F} = \frac{1}{4\pi \epsilon_o} \frac{q_1 q_2}{r^2} \hat{r}$

1 $\quad$ $\vec{F} = \frac{1}{4\pi \epsilon_o} \times \frac{q \times 2q}{(\sqrt{2}a)^2} \times \frac{(a\hat{i} + a j)}{\sqrt{2}a}$

$$\vec{F} = \frac{1}{4\pi \epsilon_o} \times \frac{2q^2}{2a^2} \times \frac{(\hat{i} + j)}{\sqrt{2}}$$

$$\vec{F} = \frac{1}{4\pi \epsilon_o} \times \frac{q^2}{\sqrt{2}a^2} \times (\hat{i} + j)$$

½ $\quad$ $\vec{F} = \frac{q^2}{4\sqrt{2}\pi \epsilon_o a^2} (\hat{i} + j)$

Note: Award 1 mark if a student calculates the magnitude of force only.

$$\left|\vec{F}\right| = \frac{1}{4\pi \epsilon_o} \frac{q^2}{a^2}$$

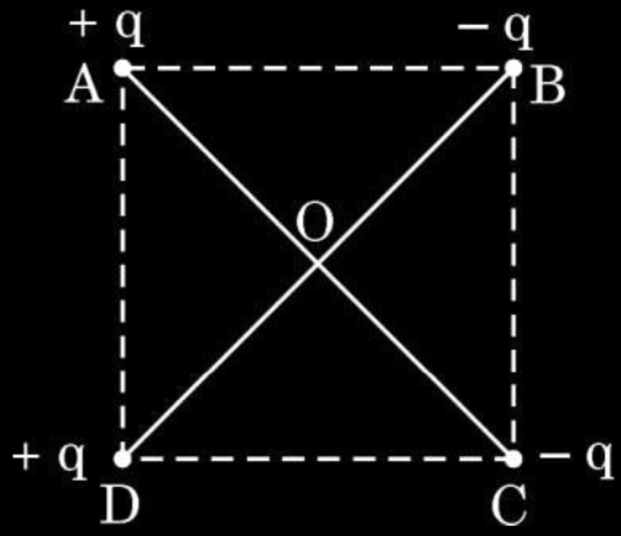
Alternatively

Give full credit if a student uses component method to solve the question.

(b) (i) Derive an expression for the electric field at a point on the equatorial plane of an electric dipole consisting of charges q and – q separated by a distance 2a.

(ii) The distance of a far off point on the equatorial plane of an electric dipole is halved. How will the electric field be affected for the dipole ?

(iii) Two identical electric dipoles are placed along the diagonals of a square ABCD of side $\sqrt{2}$ m as shown in the figure. Obtain the magnitude and direction of the net electric field at the centre (O) of the square. 5



i) Derivation of electric field	2
ii) Effect on electric field	1
iii) Finding magnitude and direction of electric field	2

i) $E_{+q} = \dfrac{q}{4\pi \in_o} \times \dfrac{1}{r^2 + a^2}$

$E_{-q} = \dfrac{q}{4\pi \in_o} \times \dfrac{1}{r^2 + a^2}$ ½

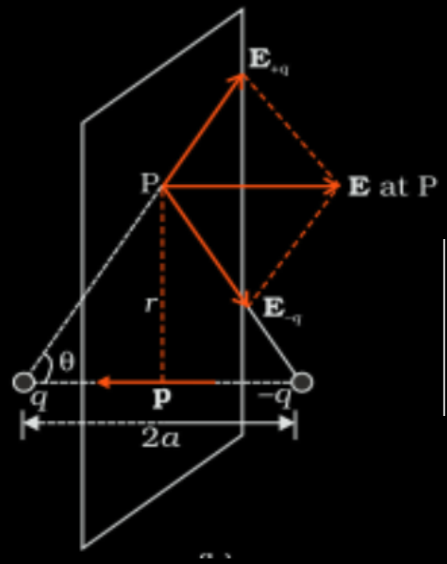
The components normal to dipole axis cancel away. The components along the dipole axis add up. ½

Total electric field is opposite to dipole moment.

$$\vec{E} = -(E_{+q} + E_{-q})\cos\theta\,\hat{p}$$

$$= \frac{-2qa}{4\pi \in_o (r^2 + a^2)^{3/2}}\hat{p}$$

$$= \frac{-\vec{p}}{4\pi \in_o (r^2 + a^2)^{3/2}}$$ ½



½

Deduct ½ mark if the expression of electric field is not in vector form.

ii) At far off point r >> a

$$\vec{E} = \frac{-\vec{p}}{4\pi \in_o r^3}$$

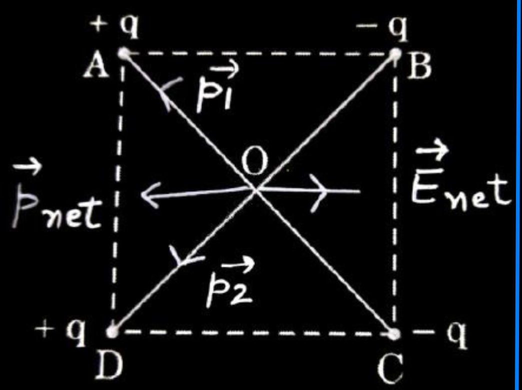
When distance is halved.

$$\vec{E} = \frac{-\vec{p}}{4\pi \in_o (\frac{r}{2})^3}$$ ½

$$= \frac{-8\vec{p}}{4\pi \in_o r^3}$$

$\vec{E}$ becomes 8 times ½

iii)



$p_1 = q \times 2Cm$ (along OA)

$p_2 = q \times 2Cm$ (along OD)

$p_{net} = \sqrt{p_1^2 + p_2^2}$

$= 2\sqrt{2}\, q\, Cm$

Electric field at centre O

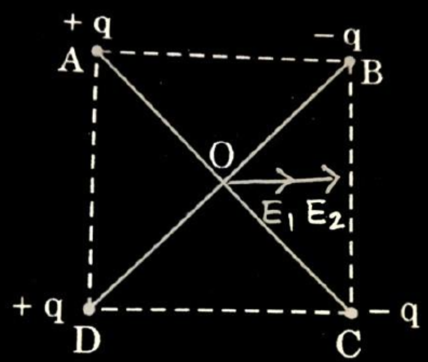
$$E = \frac{kp_{net}}{(r^2+a^2)^{3/2}}$$

at point O, r = 0 , a = 1 m

$$E = \frac{k \times 2\sqrt{2}q}{1^3} = 2\sqrt{2}kq = \frac{2\sqrt{2}q}{4\pi \in_o}$$

Along DC

Alternatively



Considering AB as dipole, electric field at O

$$E_1 = \frac{2kq \times a}{((\frac{1}{\sqrt{2}})^2 + (\frac{1}{\sqrt{2}})^2)^{3/2}} = \frac{2kqa}{(\frac{1}{2}+\frac{1}{2})^{3/2}} = 2kqa$$

Similarly considering DC as another dipole, electric field at O

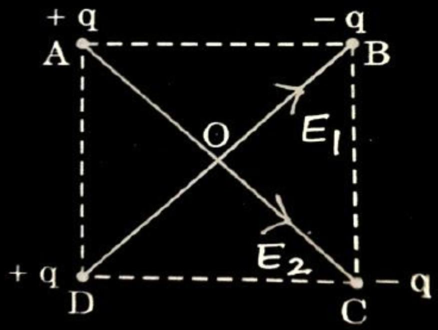
$$E_2 = \frac{2kq \times a}{((\frac{1}{\sqrt{2}})^2 + (\frac{1}{\sqrt{2}})^2)^{3/2}} = \frac{2kqa}{(\frac{1}{2}+\frac{1}{2})^{3/2}} = 2kqa$$

$$E_{net} = E_1 + E_2 = 4kqa = \frac{1}{4\pi \in_o} \times 4 \times \frac{1}{\sqrt{2}} \times q$$

$$= 2\sqrt{2}kq = \frac{2\sqrt{2}q}{4\pi \in_o}$$

Along DC

Alternatively



$$E = \frac{kq}{r^2}$$

AC= BD= 2m

r = OA =OB= OC=OD=1m

Electric field at O due to charges at B and D

$E_1 = E_B + E_D$

$$E_1 = \frac{kq}{1^2} + \frac{kq}{1^2} \quad \text{along OB}$$

$= 2kq$

Electric field at O due to charges at A and C

$E_2 = E_A + E_C$

$$E_2 = \frac{kq}{1^2} + \frac{kq}{1^2}$$

$= 2kq$ along OC

$$E_{net} = \sqrt{E_1^2 + E_2^2}$$

$$= 2\sqrt{2}\,kq = \frac{2\sqrt{2}q}{4\pi \in_o}$$

Along DC

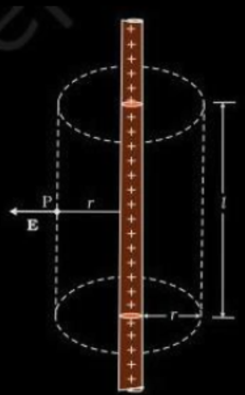
33. (a) (i) Use Gauss' law to obtain an expression for the electric field due to an infinitely long thin straight wire with uniform linear charge density λ.

(ii) An infinitely long positively charged straight wire has a linear charge density λ. An electron is revolving in a circle with a constant speed v such that the wire passes through the centre, and is perpendicular to the plane, of the circle. Find the kinetic energy of the electron in terms of magnitudes of its charge and linear charge density λ on the wire.

(iii) Draw a graph of kinetic energy as a function of linear charge density λ. **5**

(i) Derivation of the expression	-	2
(ii) Finding kinetic energy of electron	-	2
(iii)Graph	-	1

(i)



½

Flux through the Gaussian surface

$\emptyset = E.2\pi rl$

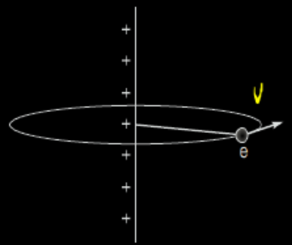
According to Gauss's law

$$E.2\pi rl = \frac{q}{\epsilon_0}$$ ½

$$q = \lambda l$$ ½

$$E = \frac{\lambda}{2\pi \epsilon_0 r}$$ ½

(i) $E = \dfrac{\lambda}{2\pi \epsilon_0 r}$



½

$$\frac{mv^2}{r} = eE$$

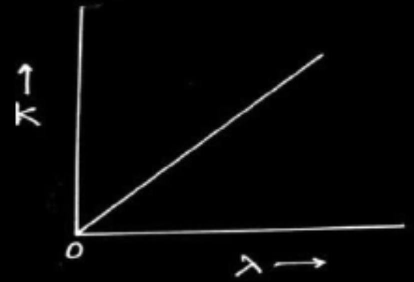
$\therefore$ Kinetic energy $K = \dfrac{1}{2}mv^2$ ½

$$= \frac{1}{2}eEr$$

$$= \frac{1}{2}e\frac{\lambda.r}{2\pi \epsilon_0 r} = \frac{e\lambda}{4\pi \epsilon_0}$$ ½

(ii) Kinetic energy $K = = \dfrac{e\lambda}{4\pi \epsilon_0}$ 1

$\therefore\ K \propto \lambda$



9. A point charge q_0 is moving along a circular path of radius a, with a point charge $-Q$ at the centre of the circle. The kinetic energy of q_0 is **1**

(a) $\dfrac{q_0 Q}{4\pi \epsilon_0 a}$

(b) $\dfrac{q_0 Q}{8\pi \epsilon_0 a}$

(c) $\dfrac{q_0 Q}{4\pi \epsilon_0 a^2}$

(d) $\dfrac{q_0 Q}{8\pi \epsilon_0 a^2}$

9. A point charge q_0 is moving along a circular path of radius a, with a point charge $-Q$ at the centre of the circle. The kinetic energy of q_0 is **1**

(a) $\frac{q_0 Q}{4\pi\in_0 a}$ (b) $\frac{q_0 Q}{8\pi\in_0 a}$

(c) $\frac{q_0 Q}{4\pi\in_0 a^2}$ (d) $\frac{q_0 Q}{8\pi\in_0 a^2}$

(b) $\frac{q_0 Q}{8\pi\varepsilon_0 a}$

7. An electric dipole of dipole moment 2×10^{-8} C-m in a uniform electric field experiences a maximum torque of 6×10^{-4} N-m. The magnitude of electric field is **1**

(a) $2.2 \times 10^{3}\, Vm^{-1}$

(b) $1.2 \times 10^{4}\, Vm^{-1}$

(c) $3.0 \times 10^{4}\, Vm^{-1}$

(d) $4.2 \times 10^{3}\, Vm^{-1}$

7. An electric dipole of dipole moment 2×10^{-8} C-m in a uniform electric field experiences a maximum torque of 6×10^{-4} N-m. The magnitude of electric field is **1**

(a) 2.2×10^3 Vm^{-1} (b) 1.2×10^4 Vm^{-1}

(c) 3.0×10^4 Vm^{-1} (d) 4.2×10^3 Vm^{-1}

(c) 3×10^4 Vm^{-1}

5. An isolated point charge particle produces an electric field $\vec{E}$ at a point 3 m away from it. The distance of the point at which the field is $\frac{\vec{E}}{4}$ will be **1**

(a) 2 m (b) 3 m

(c) 4 m (d) 6 m

5. An isolated point charge particle produces an electric field $\vec{E}$ at a point 3 m away from it. The distance of the point at which the field is $\frac{\vec{E}}{4}$ will be **1**

(a) 2 m

(b) 3 m

(c) 4 m

(d) 6 m

(d) 6 m

1. The magnitude of the electric field due to a point charge object at a distance of 4.0 m is 9 N/C. From the same charged object the electric field of magnitude, 16 $\frac{\text{N}}{\text{C}}$ will be at a distance of **1**

(a) 1 m (b) 2 m

(c) 3 m (d) 6 m

(c)
3 m

1. An electric dipole of length 2 cm is placed at an angle of 30° with an electric field 2×10^5 N/C. If the dipole experiences a torque of 8×10^{-3} Nm, the magnitude of either charge of the dipole, is **1**

(A) 4 μC (B) 7 μC

(C) 8 mC (D) 2 mC

1. An electric dipole of length 2 cm is placed at an angle of 30° with an electric field 2×10^5 N/C. If the dipole experiences a torque of 8×10^{-3} Nm, the magnitude of either charge of the dipole, is 1

(A) 4 μC (B) 7 μC

(C) 8 mC (D) 2 mC

(A) 4μC

1. A charge Q is placed at the centre of a cube. The electric flux through one if its face is 1

(A) $\frac{Q}{\varepsilon_0}$

(B) $\frac{Q}{6\varepsilon_0}$

(C) $\frac{Q}{8\varepsilon_0}$

(D) $\frac{Q}{3\varepsilon_0}$

1. A charge Q is placed at the centre of a cube. The electric flux through one if its face is **1**

(A) $\frac{Q}{\varepsilon_0}$

(B) $\frac{Q}{6\varepsilon_0}$

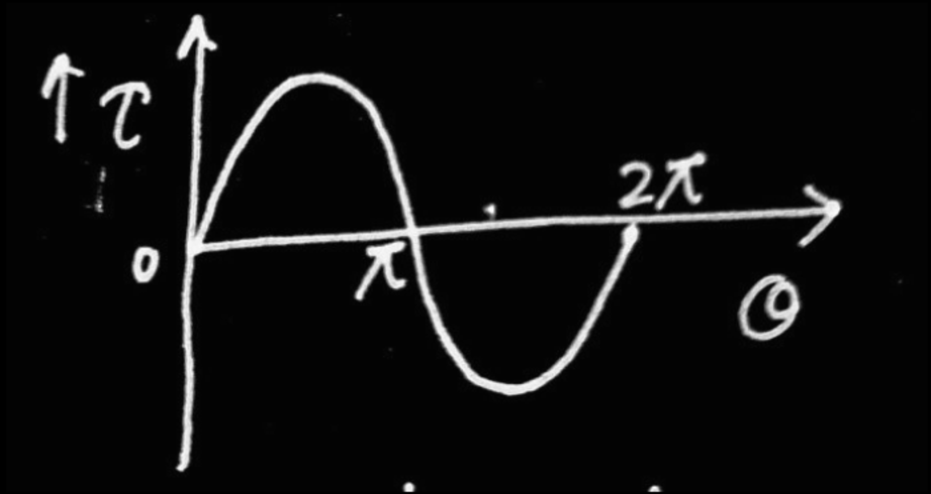
(C) $\frac{Q}{8\varepsilon_0}$

(D) $\frac{Q}{3\varepsilon_0}$

(B) $\frac{Q}{6 \in_o}$

22. An electric dipole of dipole moment ($\vec{p}$) is kept in a uniform electric field $\vec{E}$. Show graphically the variation of torque acting on the dipole (τ) with its orientation (θ) in the field. Find the orientation in which torque is (i) zero and (ii) maximum. **2**

(a) Graphical representation of torque - θ	(1)
Finding the orientation for	
(i) Zero torque	(½)
(ii) Maximum torque	(½)

	1
(i) Torque is zero for orientation corresponding to $\theta = 0^0$ and $\theta = 180^0$	½
(ii) Torque is maximum for orientation corresponding to $\theta = \frac{\pi}{2}$ and $\frac{3\pi}{2}$	½
(Note: Award full credit of 1 mark for parts (i) and (ii) if a student writes only one value of angle in each case.)	

31. (a) (i) Why do we use a small test charge to measure an electric field ?

(ii) A small stationary positively charged particle is free to move in an electric field. In which direction will it begin to move ?

(iii) Two point charges Q_1 (40 μC) and Q_2 (–16 μC) are placed along x-axis at 0 cm and 24 cm from the origin respectively. Calculate the net force on a third charge Q_3 (– 2·5 μC) placed at x = 36 cm from the origin. *5*

(i) Reason for using small test charge	1
(ii) Identifying the direction of force	1
(iii) Calculating the net force	3

(i) So that it does not disturb the distribution of charges whose electric field is to be measured. Otherwise field will be different from the actual field. 1

(ii) It will move in the direction of electric field. 1

(iii) $F = F_2 - F_1$

$$= \frac{kQ_2Q_3}{r_{23}^{\;2}} - \frac{kQ_1Q_3}{r_{13}^{\;2}}$$ 1

$$F = \frac{9\times10^9\times16\times2.5\times10^{-12}}{(12\times10^{-2})^2} - \frac{9\times10^9\times40\times2.5\times10^{-12}}{(36\times10^{-2})^2}$$

$$F = 25 - 6.94$$

$$F = 18.06\,N$$ 1

2. 10^9 electrons are transferred to a pith ball with charge $0{\cdot}16$ nC. Its charge now is :

(a) Zero

(b) $-3{\cdot}2 \times 10^{-10}$ C

(c) $-1{\cdot}6 \times 10^{-9}$ C

(d) $3{\cdot}2 \times 10^{-10}$ C

2. 10^9 electrons are transferred to a pith ball with charge $0 \cdot 16$ nC. Its charge now is :

(a) Zero

(b) $-3 \cdot 2 \times 10^{-10}$ C

(c) $-1 \cdot 6 \times 10^{-9}$ C

(d) $3 \cdot 2 \times 10^{-10}$ C

(a) Zero

1. The electric flux through a Gaussian spherical surface enclosing a point charge q is ϕ. If the charge is replaced by an electric dipole, magnitude of its dipole moment being 2qa, the flux through the surface will be :

(a) $2\,\phi$

(b) ϕ

(c) $\frac{\phi}{2}$

(d) Zero

1. The electric flux through a Gaussian spherical surface enclosing a point charge q is ϕ. If the charge is replaced by an electric dipole, magnitude of its dipole moment being 2qa, the flux through the surface will be :

(a) 2ϕ

(b) ϕ

(c) $\frac{\phi}{2}$

(d) Zero

(d) Zero

22. A uniform electric field is represented as $\vec{E} = (3 \times 10^3 \frac{N}{C})\ \hat{i}$. Find the electric flux of this field through a square of side 10 cm when the :

(a) plane of the square is parallel to y-z plane, and

(b) the normal to plane of the square makes an angle of 60° with the x-axis. *2*

(a) Calculation of electric flux	½ + ½
(b) Calculation of electric flux	½ + ½

(a)

$$\Phi = \vec{E}.A\hat{i} = EA$$ ½

$$= 3\times10^{3}\times(100\times10^{-4})$$

$$= 30NC^{-1}m^{2}$$ ½

(b)

$$\Phi = EA\cos 60^{o}$$ ½

$$= 15NC^{-1}m^{2}$$ ½

1. An electric dipole with dipole moment $\vec{P} = P_0\hat{i} - P_0\hat{j}$ is placed in an electric field $\vec{E} = E_1\hat{i} + E_2\hat{j}$, where P_0, E_1 and E_2 are constants. The torque $\vec{\tau}$ acting on the dipole is :

(a) $P_0(E_2 - E_1)\hat{k}$

(b) $P_0(E_2 + E_1)\hat{k}$

(c) $-P_0(E_2 + E_1)\hat{k}$

(d) $P_0(E_1 - E_2)\hat{k}$

1. An electric dipole with dipole moment $\vec{P} = P_0\hat{i} - P_0\hat{j}$ is placed in an electric field $\vec{E} = E_1\hat{i} + E_2\hat{j}$, where P_0, E_1 and E_2 are constants. The torque $\vec{\tau}$ acting on the dipole is :

(a) $P_0(E_2 - E_1)\hat{k}$ (b) $P_0(E_2 + E_1)\hat{k}$

(c) $-P_0(E_2 + E_1)\hat{k}$ (d) $P_0(E_1 - E_2)\hat{k}$

(b) $p_0(E_2 + E_1)\,k$

22. The inward and the outward electric flux through a Gaussian surface are 2ϕ and ϕ respectively.

(a) What is the net charge enclosed by the surface ?

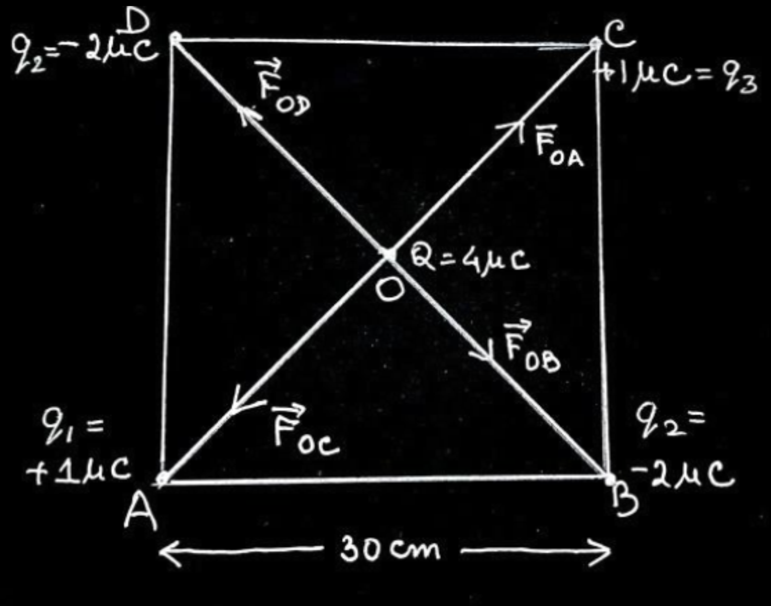
(b) If the net outward flux through the surface were zero, can it be concluded that there were no charges inside the surface ? Justify your answer. *2*

(a)	Calculation of net charge	1
(b)	Conclusion	½
	Justification	½

(a) Net outward flux $= -2\,\Phi + \Phi$	
$= -\Phi$	½
Charge enclosed $= -\Phi\, \varepsilon_o$	½
(b) No	½
There may be charges (positive and negative) and net charge is zero.	½

18. (a) Four point charges of 1 μC, –2 μC, 1 μC and –2 μC are placed at the corners A, B, C and D respectively, of a square of side 30 cm. Find the net force acting on a charge of 4 μC placed at the centre of the square.

Diagram showing direction of forces	1
Finding net force	1



1

OA = OB = OC = OD = r

Net force on charge $4\mu C$

$$\vec{F} = \vec{F}_{OA} + \vec{F}_{OB} + \vec{F}_{OC} + \vec{F}_{OD}$$ ½

$$\vec{F}_{OA} = -\vec{F}_{OC} \quad \Rightarrow \vec{F}_{OA} + \vec{F}_{OC} = 0$$

$$\vec{F}_{OB} = -\vec{F}_{OD} \quad \Rightarrow \vec{F}_{OB} + \vec{F}_{OD} = 0$$ ½

$$\vec{F} = 0$$

Alternatively

$$F_{OA} = F_{OC} = \frac{9\times10^{9}\times4\times10^{-6}\times1\times10^{-6}}{(15\sqrt{2}\times10^{-2})^{2}}$$

$$= 0.8\,N$$ ½

$$F_{OB} = F_{OD} = 1.6\ N$$

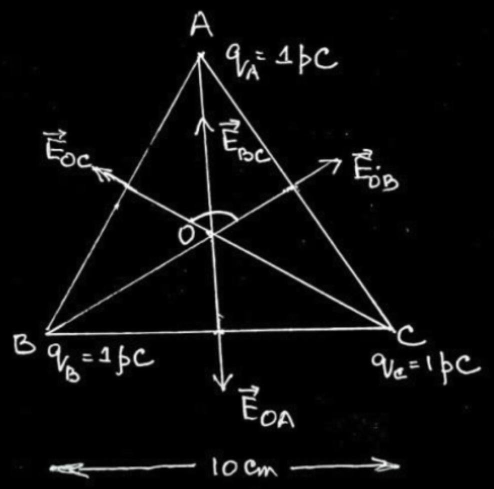
$$F_1 = F_{OA} - F_{OC} = 0$$

$$F_2 = F_{OB} - F_{OD} = 0$$ ½

Net Force $F = 0$

18. (b) Three point charges, 1 pC each, are kept at the vertices of an equilateral triangle of side 10 cm. Find the net electric field at the centroid of triangle.

Finding net electric field at centroid	2



1

$$q_A = q_B = q_C = 1pC$$

$$AO = BO = CO = r$$

$$\left|\vec{E}_{OA}\right| = \left|\vec{E}_{OB}\right| = \left|\vec{E}_{OC}\right|$$

$$\vec{E}_{BC} = \vec{E}_{OB} + \vec{E}_{OC}$$ ½

$$E_{BC} = \sqrt{E_{OB}^2 + E_{OC}^2 + 2E_{OB}E_{OC}\cos 120^0}$$

$$E_{BC} = E_{OB} \quad , \quad \vec{E}_{OA} = -\vec{E}_{BC}$$

Net electric field $\vec{E}_O = \vec{E}_{OA} + \vec{E}_{BC}$ ½

$$\vec{E}_O = 0$$

Alternatively

$$E_{OA} = E_{OB} = E_{OC} = 2.7 \ NC^{-1}$$

$$E_{BC} = \sqrt{E_{OB}^2 + E_{OC}^2 + 2E_{OB}E_{OC}\cos 120^0}$$ ½

$$= E_{OB}$$

$$As \quad \vec{E}_{BC} = -\vec{E}_{OA}$$

$$\vec{E}_{BC} + \vec{E}_{OA} = 0$$ ½

Net electric field is zero.

Alternatively

$$\left|\vec{E}_{OA}\right| = \left|\vec{E}_{OB}\right| = \left|\vec{E}_{OC}\right|$$

Electric field vectors are making an angle of 120^0 with each other. They make a closed polygon. So vector sum of all electric field vectors will be zero.

$$\vec{E} = 0$$ 2

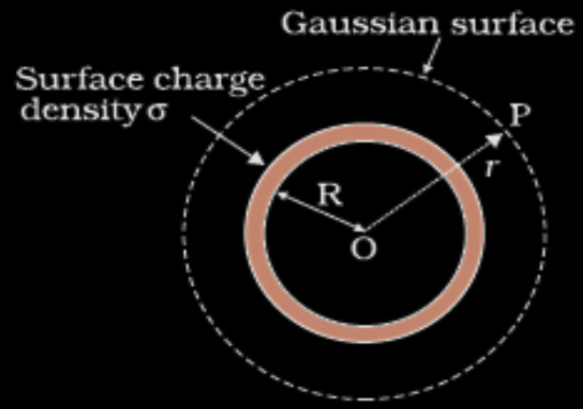
33. (b) (i) A thin spherical shell of radius R has a uniform surface charge density σ. Using Gauss' law, deduce an expression for electric field (i) outside and (ii) inside the shell.

(ii) Two long straight thin wires AB and CD have linear charge densities 10 μC/m and –20 μC/m, respectively. They are kept parallel to each other at a distance 1 m. Find magnitude and direction of the net electric field at a point midway between them.

(i) Deduction of an expression for electric field for (i) and (ii)	3
(ii) Finding magnitude and direction of the net electric field	2

(i)

(i) **Electric Field outside the shell**



½

Electric flux through Gaussian surface

$\Phi = E \times 4\pi r^2$

Charge enclosed by the Gaussian surface

$Q = \sigma \times 4\pi R^2$

Using Gauss' law: $\int \vec{E}.\vec{ds} = \dfrac{Q}{\varepsilon_0}$ ½

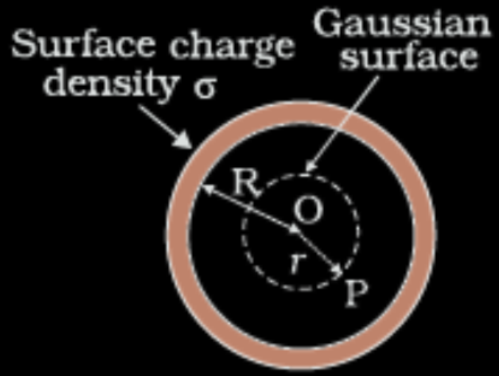
$$E \times 4\pi r^2 = \frac{(\sigma 4\pi R^2)}{\varepsilon_0}$$

$$\therefore\ E = \frac{\sigma R^2}{\varepsilon_0 r^2}$$

½

$$\vec{E} = \frac{\sigma R^2}{\varepsilon_0 r^2}\hat{r}$$

(ii) **Field inside the shell**



½

Electric flux through Gaussian surface

$\Phi = E \times 4\pi r^2 \qquad (\because r < R)$

Charge enclosed by the Gaussian surface

$Q = 0$

By Gauss' Law

$E \times 4\pi r^2 = 0$ ½

i.e. $E = 0$ ½

(Note: Award full credit of this part if a student writes directly E=0, mentioning as there is no charge enclosed by Gaussian surface)

(ii) Electric field due to a long straight charged wire of linear charged density λ

$$E = \frac{\lambda}{2\pi\varepsilon_0 r}$$ ½

Net electric field at the mid-point

$E_{net} = E_1 + E_2$

$$= \frac{\lambda_1}{2\pi\varepsilon_0 r} + \frac{\lambda_2}{2\pi\varepsilon_0 r}$$ ½

$$E_{net} = \frac{1}{2\pi\varepsilon_0 r}\left[\lambda_1 + \lambda_2\right]$$

$$= \frac{2\times 9\times 10^9}{0.5}\left[10+20\right]\times 10^{-6}$$

$= 1.08 \times 10^6\ NC^{-1}$ ½

$\vec{E}_{net}$ is directed towards CD. ½

23. (a) Define the term 'electric flux' and write its dimensions.

(b) A plane surface, in shape of a square of side 1 cm is placed in an electric field $\vec{E} = \left(100\,\frac{N}{C}\right)\hat{i}$ such that the unit vector normal to the surface is given by $\hat{n} = 0.8\hat{i} + 0.6\hat{k}$. Find the electric flux through the surface.

(a) Defining the term electric flux	1
Writing dimensions	½
(b) Finding the electric flux	1½

(a) It is the measure of the total number of electric field lines passing through a surface normally.	1
Alternatively	
Surface integral of electric field over a surface.	
Alternatively	
$\phi_E = \vec{E} \cdot \vec{A}$	
$\left[ML^3T^{-3}A^{-1}\right]$	½
(b) $\phi_E = \vec{E}.\vec{A}$	½
$= (100\hat{i}).(10^{-4}\ \hat{n})$	
$= (100\hat{i}).(0.8\hat{i} + 0.6\hat{k})10^{-4}$	½
$= 8 \times 10^{-3}\ \mathrm{Nm^2C^{-1}}$	½

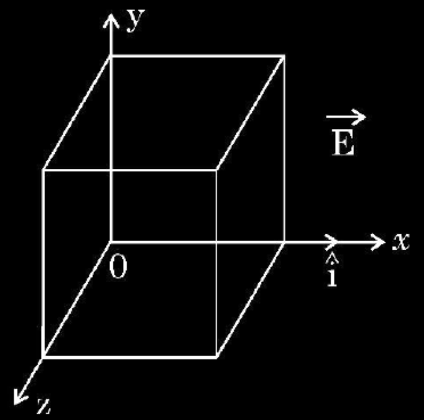
(b) (i) Using Gauss's law, show that the electric field $\vec{E}$ at a point due to a uniformly charged infinite plane sheet is given by $\vec{E} = \frac{\sigma}{2\varepsilon_0}\hat{n}$ where symbols have their usual meanings.

(ii) Electric field $\vec{E}$ in a region is given by

$\vec{E} = (5x^2 + 2)\hat{i}$

where E is in N/C and x is in meters.

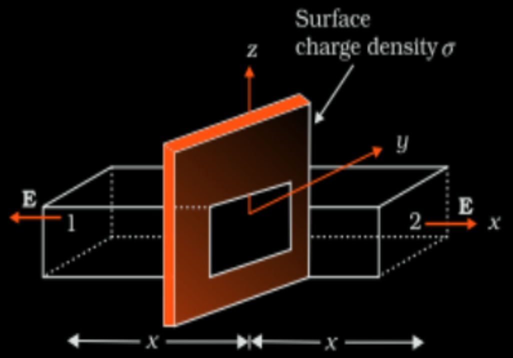
A cube of side 10 cm is placed in the region as shown in figure.



Calculate (1) the electric flux through the cube, and (2) the net charge enclosed by the cube. **5**

i) Showing electric field at a point due to a uniformly charged infinite plane sheet		3
ii) Calculating	(1) electric flux through the cube	1
	(2) charge enclosed by cube	1

(i)



1

$$\oint \vec{E}.d\vec{s} = \int_1 \vec{E}.d\vec{s} + \int_2 \vec{E}.d\vec{s}$$ ½

$$= 2EA$$

From Gauss's law

$$\oint \vec{E}.d\vec{s} = \frac{q}{\varepsilon_0}$$ ½

$$2EA = \frac{\sigma A}{\varepsilon_0}$$

$$E = \frac{\sigma}{2\varepsilon_0}$$ ½

Vectorially $\vec{E} = \frac{\sigma}{2\varepsilon_0}\hat{n}$ ½

Electric field is normally outward of the sheet.

(ii) (1) Electric flux through the cube

$$\phi = \phi_L + \phi_R$$ ½

$$\phi = \int \vec{E}_L.d\vec{s} + \int \vec{E}_R.d\vec{s}$$

$$= -2\times100\times10^{-4} + [5\times(10\times10^{-2})^2 + 2]\times100\times10^{-4}$$

$$\phi = 5\times10^{-4}\ Nm^2C^{-1}$$ ½

(2)

$$\phi = \frac{q_{en}}{\varepsilon_0}$$ ½

$$q_{en} = \phi.\varepsilon_0$$

$$= 5\times10^{-4}\times8.85\times10^{-12}$$

$$= 4.43\times10^{-15}C$$ ½

2. An infinite long straight wire having a charge density λ is kept along y′y axis in x-y plane. The Coulomb force on a point charge q at a point P (x, 0) will be **1**

(A) attractive and $\dfrac{q\lambda}{2\pi\varepsilon_0 x}$

(B) repulsive and $\dfrac{q\lambda}{2\pi\varepsilon_0 x}$

(C) attractive and $\dfrac{q\lambda}{\pi\varepsilon_0 x}$

(D) repulsive and $\dfrac{q\lambda}{\pi\varepsilon_0 x}$

2. An infinite long straight wire having a charge density λ is kept along $y'y$ axis in x-y plane. The Coulomb force on a point charge q at a point P (x, 0) will be **1**

(A) attractive and $\frac{q\lambda}{2\pi\varepsilon_0 x}$

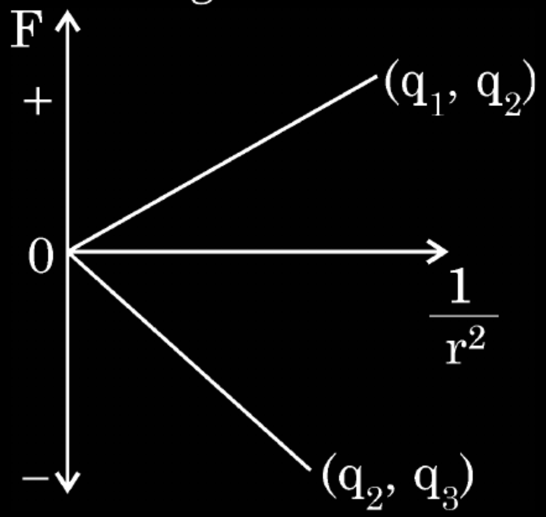
(B) repulsive and $\frac{q\lambda}{2\pi\varepsilon_0 x}$

(C) attractive and $\frac{q\lambda}{\pi\varepsilon_0 x}$

(D) repulsive and $\frac{q\lambda}{\pi\varepsilon_0 x}$

(B) Repulsive and $\frac{q\lambda}{2\pi\varepsilon_0 x}$

2. The Coulomb force (F) versus ($1/r^2$) graphs for two pairs of point charges (q_1 and q_2) and (q_2 and q_3) are shown in figure. The charge q_2 is positive and has least magnitude. Then **1**



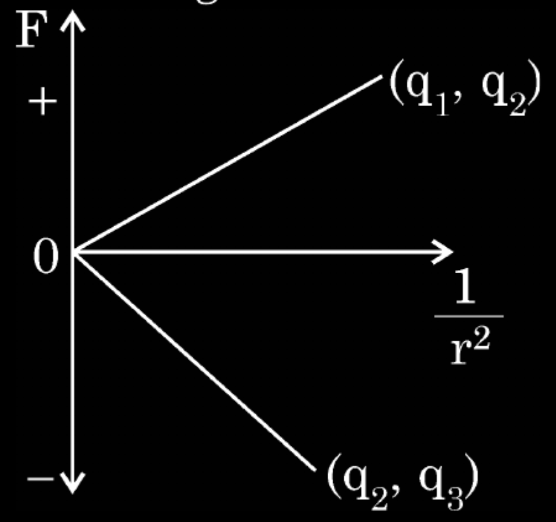
(A) $q_1 > q_2 > q_3$

(B) $q_1 > q_3 > q_2$

(C) $q_3 > q_2 > q_1$

(D) $q_3 > q_1 > q_2$

2. The Coulomb force (F) versus ($1/r^2$) graphs for two pairs of point charges (q_1 and q_2) and (q_2 and q_3) are shown in figure. The charge q_2 is positive and has least magnitude. Then **1**



(A) $q_1 > q_2 > q_3$ (B) $q_1 > q_3 > q_2$

(C) $q_3 > q_2 > q_1$ (D) $q_3 > q_1 > q_2$

(D) $q_3 > q_1 > q_2$

2. Two identical small conducting balls B_1 and B_2 are given –7 pC and + 4 pC charges respectively. They are brought in contact with a third identical ball B_3 and then separated. If the final charge on each ball is –2 pC, the initial charge on B_3 was **1**

(A) –2 pC

(B) –3 pC

(C) –5 pC

(D) –15 pC

2. Two identical small conducting balls B_1 and B_2 are given –7 pC and + 4 pC charges respectively. They are brought in contact with a third identical ball B_3 and then separated. If the final charge on each ball is –2 pC, the initial charge on B_3 was **1**

(A) –2 pC (B) –3 pC

(C) –5 pC (D) –15 pC

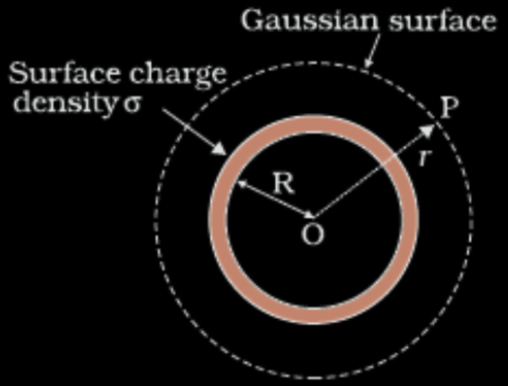
(B) -3 pC

(b) (i) A charge + Q is placed on a thin conducting spherical shell of radius R. Use Gauss's theorem to derive an expression for the electric field at a point lying (i) inside and (ii) outside the shell.

(ii) Show that the electric field for same charge density (σ) is twice in case of a conducting plate or surface than in a nonconducting sheet. *5*

(i) Expression for electric field at appoint lying	
(i) inside	1
(ii) outside	2
(ii) Explanation	2

(i) **Field inside the shell**



The Flux through the Gaussian surface is

$= E \times 4\pi R^2$ ½

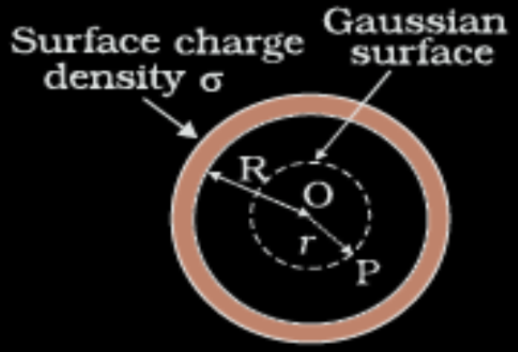
In this case Gaussian surface enclosed no charge.

Hence $E \times 4\pi R^2 = 0$

$E = 0$ ½

(Note: Award full credit of this part if a student writes directly E=0, mentioning as there is no charge enclosed by Gaussian surface)

(ii) **Field outside the shell**-



½

Electric flux through Gaussian surface

$$E \times 4\pi r^2 = \frac{(\sigma\, 4\pi R^2)}{\varepsilon_0}$$ ½

Charge enclosed by the Gaussian surface

$$E \times 4\pi r^2 = \frac{(\sigma\, 4\pi R^2)}{\varepsilon_0}$$

Using Gauss's law:

$$\int \vec{E}.\overrightarrow{ds} = \frac{Q}{\varepsilon_0}$$ ½

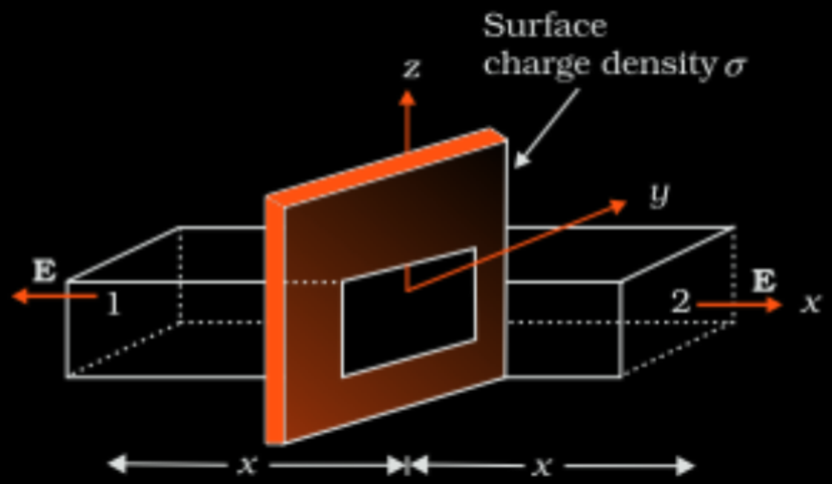
$$E \times 4\pi r^2 = \frac{(\sigma\, 4\pi R^2)}{\varepsilon_0}$$

$$E = \frac{\sigma}{\varepsilon_0}\frac{R^2}{r^2} = \frac{q}{4\pi\varepsilon_0 r^2}$$ ½

(ii) For conducting sheet,

Electric field due to a conducting sheet

$$E_c = \frac{\sigma}{\varepsilon_0}$$ ½



For non-conducting sheet

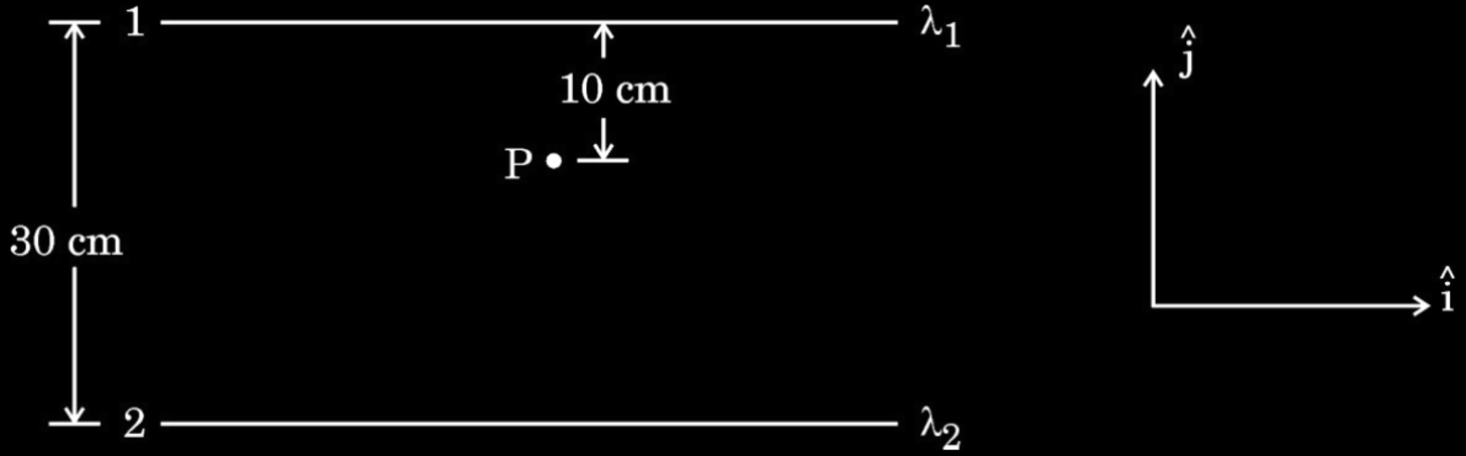
$$E_{nc} = \frac{\sigma}{2\varepsilon_0}$$ ½

Since surface charge density is same. ½

$$2E_{nc} = E_c$$ ½

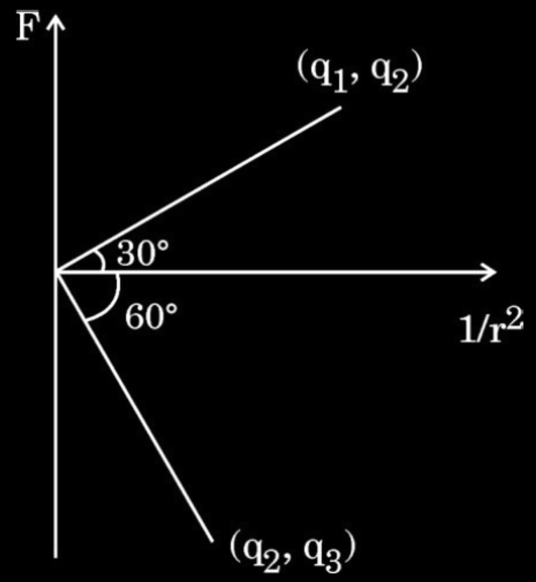
(b) (i) State Gauss's Law in electrostatics. Apply this to obtain the electric field $\vec{E}$ at a point near a uniformly charged infinite plane sheet.

(ii) Two long straight wires 1 and 2 are kept as shown in the figure. The linear charge density of the two wires are $\lambda_1 = 10\ \mu C/m$ and $\lambda_2 = -20\ \mu C/m$. Find the net force $\vec{F}$ experienced by an electron held at point P. *5*



(i) Statement of Gauss's Law	1
Obtaining expression for electric field	2
(ii) Finding net force on electron	2

1. Coulomb force F versus $\left(\frac{1}{r^2}\right)$ graphs for two pairs of point charges (q_1 and q_2) and (q_2 and q_3) are shown in the figure. The ratio of charges $\left(\frac{q_1}{q_3}\right)$ is :



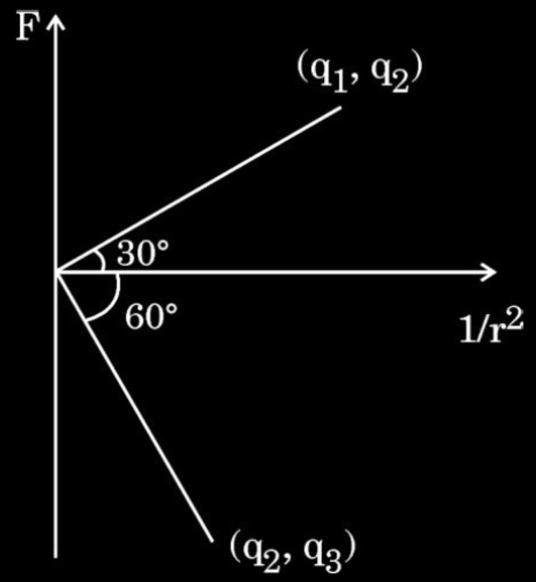
(A) $\sqrt{3}$

(B) $\frac{1}{\sqrt{3}}$

(C) 3

(D) $\frac{1}{3}$

1. Coulomb force F versus $\left(\frac{1}{r^2}\right)$ graphs for two pairs of point charges (q_1 and q_2) and (q_2 and q_3) are shown in the figure. The ratio of charges $\left(\frac{q_1}{q_3}\right)$ is :



(A) $\sqrt{3}$

(B) $\frac{1}{\sqrt{3}}$

(C) 3

(D) $\frac{1}{3}$

(D) $\frac{1}{3}$

1. Two charged particles P and Q, having the same charge but different masses m_P and m_Q, start from rest and travel equal distances in a uniform electric field $\vec{E}$ in time t_P and t_Q respectively. Neglecting the effect of gravity, the ratio $\left(\frac{t_P}{t_Q}\right)$ is :

(A) $\frac{m_P}{m_Q}$

(B) $\frac{m_Q}{m_P}$

(C) $\sqrt{\frac{m_P}{m_Q}}$

(D) $\sqrt{\frac{m_Q}{m_P}}$

1. Two charged particles P and Q, having the same charge but different masses m_P and m_Q, start from rest and travel equal distances in a uniform electric field $\vec{E}$ in time t_P and t_Q respectively. Neglecting the effect of gravity, the ratio $\left(\frac{t_P}{t_Q}\right)$ is :

(A) $\frac{m_P}{m_Q}$

(B) $\frac{m_Q}{m_P}$

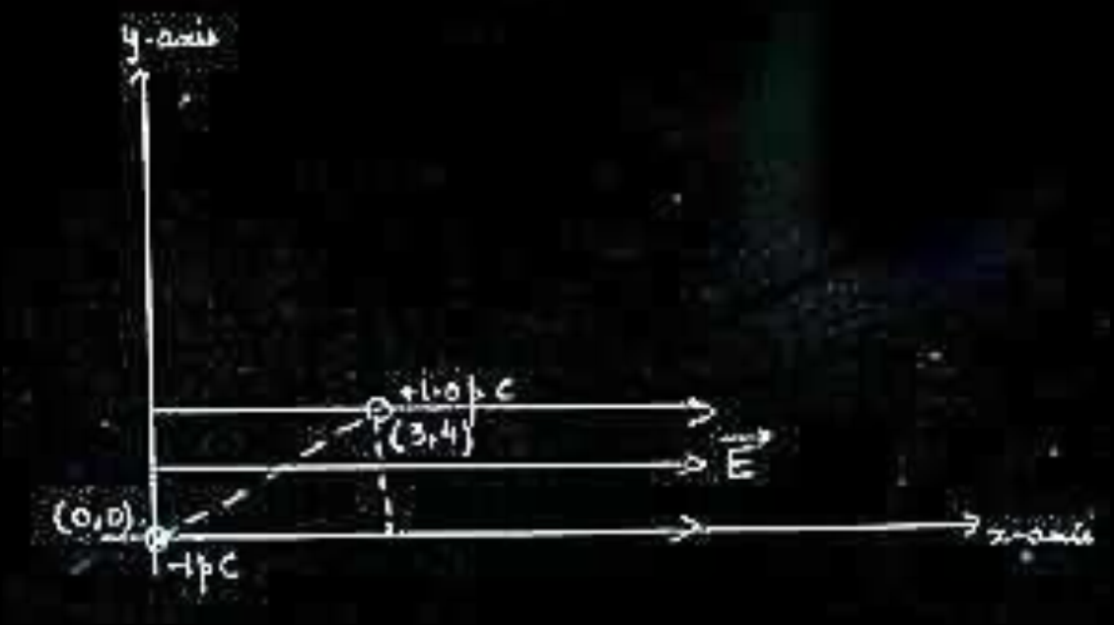
(C) $\sqrt{\frac{m_P}{m_Q}}$

(D) $\sqrt{\frac{m_Q}{m_P}}$

(C) $\sqrt{\frac{m_p}{m_Q}}$

(ii) An electric dipole consists of point charges – 1·0 pC and + 1·0 pC located at (0, 0) and (3 mm, 4 mm) respectively in x – y plane. An electric field $\vec{E} = \left(\frac{1000 \text{ V}}{\text{m}}\right)\hat{i}$ is switched on in the region. Find the torque $\vec{\tau}$ acting on the dipole.

(ii)



½

$\tau = pE\sin\theta$ ½

$= (2aq)E\sin\theta$

$= (5\times10^{-3}\times1\times10^{-12})10^{3}\times\frac{4}{5}$ ½

$= 4\times10^{-12}\ \ Nm$ ½

Direction is along –ve Z direction.

1. A thin plastic rod is bent into a circular ring of radius R. It is uniformly charged with charge density λ. The magnitude of the electric field at its centre is :

(A) $\frac{\lambda}{2\varepsilon_0 R}$ (B) Zero (C) $\frac{\lambda}{4\pi\varepsilon_0 R}$ (D) $\frac{\lambda}{4\varepsilon_0 R}$

1. A thin plastic rod is bent into a circular ring of radius R. It is uniformly charged with charge density λ. The magnitude of the electric field at its centre is :

(A) $\frac{\lambda}{2\varepsilon_0 R}$ (B) Zero (C) $\frac{\lambda}{4\pi\varepsilon_0 R}$ (D) $\frac{\lambda}{4\varepsilon_0 R}$

(B) Zero

2. A charged sphere of radius r has surface charge density σ. The electric field on its surface is E. If the radius of the sphere is doubled, keeping charge density the same, the ratio of the electric field on the old sphere to that on the new sphere will be :

(A) 1 (B) $\frac{1}{2}$ (C) $\frac{1}{4}$ (D) 4

2. A charged sphere of radius r has surface charge density σ. The electric field on its surface is E. If the radius of the sphere is doubled, keeping charge density the same, the ratio of the electric field on the old sphere to that on the new sphere will be :

(A) 1 (B) $\frac{1}{2}$ (C) $\frac{1}{4}$ (D) 4

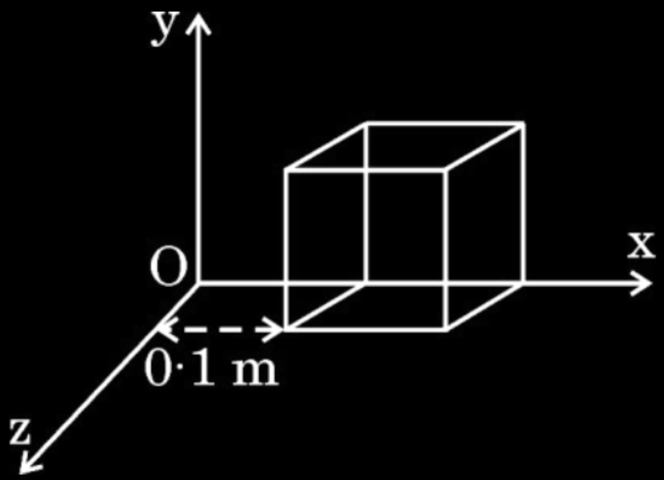
(A) 1

22. A cube of side 0·1 m is placed, as shown in the figure, in a region where electric field $\vec{E} = 500\,x\,\hat{i}$ exists. Here x is in meters and E in NC^{-1}.

Calculate : *3*

(a) the flux passing through the cube, and

(b) the charge within the cube.



Calculating	
(a) the flux passing through the cube	2
(b) the charge within the cube	1

a) $\varphi_L = \overrightarrow{E_L} \cdot \vec{A} = -[500 \times 0.1] \times [(0.1)^2] = -0.5 \text{ N m}^2 \text{ C}^{-1}$ ½

$\varphi_R = \overrightarrow{E_R} \cdot \vec{A} = [500 \times 0.2] \times [(0.1)^2] = 1 \text{ N m}^2 \text{ C}^{-1}$ ½

Net flux $= \varphi_L + \varphi_R = 0.5 \text{ N m}^2 \text{ C}^{-1}$ 1

b) flux, $\varphi = \dfrac{q}{\varepsilon_o}$ ½

charge, $q = \varphi \times \varepsilon_o$

$= 0.5\ \varepsilon_o$ ½

$= 4.4 \times 10^{-12}$ C

30. An electric dipole is a pair of equal and opposite point charges + q and – q separated by a distance 2a. A dipole can be characterised by dipole moment determined by its charge and separation between the two charges. It is a vector quantity. By convention the direction of a dipole is taken to be the direction from – q to + q. Like fields due to a point charge, fields due to a given dipole at a point can be found.

When a dipole is placed in a uniform external electric field, it experiences a torque. This torque tends to rotate the dipole and work is done in this process. This work is stored as the potential energy of the system.

(i) An electric dipole is kept in a uniform electric field such that the dipole moment is not along the electric field. It will experience : *1*

(A) A net force and a torque

(B) A net force but no torque

(C) A torque but no net force

(D) neither a net force nor a torque

(ii) An electric dipole consists of charges + q and – q, separated by distance 2a. The force acting on a point charge placed at a distance x (>> a) on the axis of the dipole is F. If the distance of the charge is doubled, the force will become : *1*

(A) $\frac{F}{16}$ (B) $\frac{F}{8}$

(C) $\frac{F}{4}$ (D) $\frac{F}{2}$

(iii) Two identical small electric dipoles, each of dipole moment $\vec{p}$ are arranged perpendicular to each other such that their negative charges coincide. The magnitude of the net dipole moment of the arrangement will be : *1*

(A) p (B) 2p

(C) $p\sqrt{2}$ (D) $\frac{p}{\sqrt{2}}$

(iv) (a) A molecule with dipole moment $3 \cdot 0 \times 10^{-30}$ C-m is placed in a uniform electric field of 2×10^{4} NC^{-1}. The maximum torque experienced by the molecule is : 1

(A) $1 \cdot 5 \times 10^{-34}$ N-m

(B) $6 \cdot 0 \times 10^{-26}$ N-m

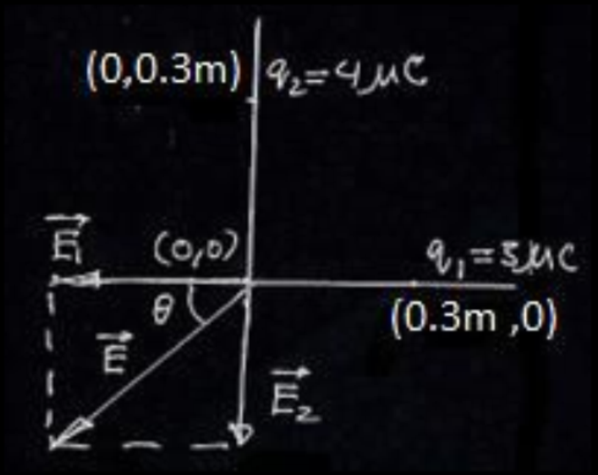
(C) $4 \cdot 5 \times 10^{34}$ N-m

(D) $6 \cdot 0 \times 10^{-34}$ N-m

30 (i)	(C)	A torque but no net force	1
(ii)	(B)	F/8	1
(iii)	(C)	$p\sqrt{2}$	1
(iv)	(a) (B)	$6\times10^{-26}\ Nm$	1

(b) Two point charges of 3 μC and 4 μC are kept in air at (0·3 m, 0) and (0, 0.3 m) in x-y plane. Find the magnitude and direction of the net electric field produced at the origin (0, 0). *2*

Finding the magnitude of electric field	1 ½
Finding the direction of electric field	½



$$\vec{E}_1 = \frac{kq_1}{r_1^2}(-\hat{i}) = \frac{9\times10^9\times3\times10^{-6}}{(0.3)^2}(-\hat{i}) = 3\times10^5(-\hat{i})\,\mathrm{NC}^{-1}$$ ½

$$\vec{E}_2 = \frac{kq_2}{r_2^2}(-\hat{j}) = \frac{9\times10^9\times4\times10^{-6}}{(0.3)^2}(-\hat{j}) = 4\times10^5(-\hat{j})\,\mathrm{NC}^{-1}$$ ½

$$\vec{E} = \vec{E}_1 + \vec{E}_2$$

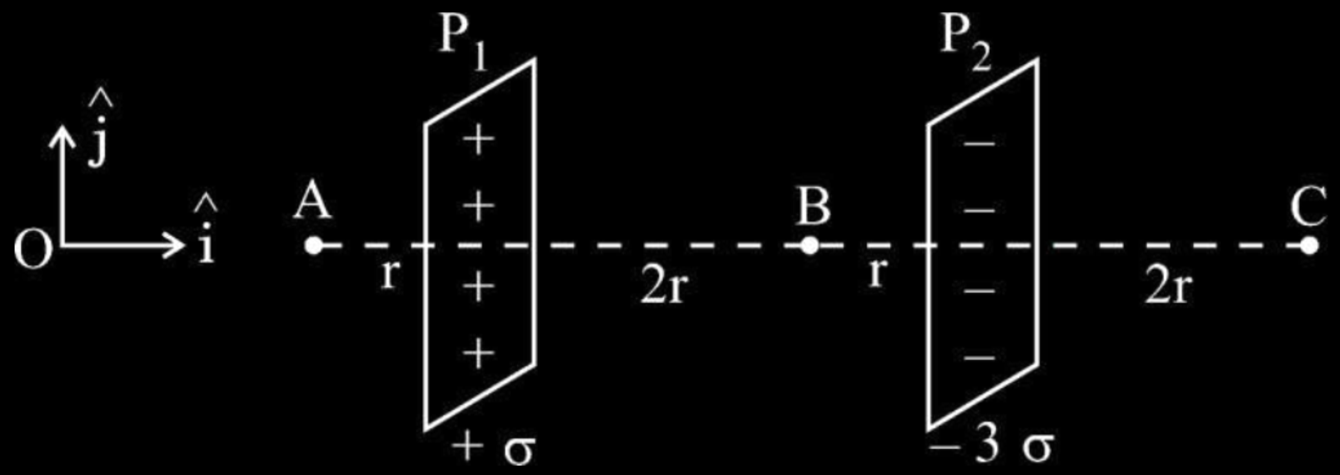
$$E = \sqrt{E_1^2 + E_2^2}$$

$$E = 5\times10^5\ \mathrm{NC}^{-1}$$ ½

$$\tan\theta = \frac{4}{3}$$

$\theta = \tan^{-1}\left(\frac{4}{3}\right)$ inclination with respect to the x-axis (in III quadrant). ½

(b) Two large plane sheets P_1 and P_2 having charge densities $+\sigma$ and -3σ respectively are arranged parallel to each other as shown in the figure. Find the net electric field ($\vec{E}$) at points A, B and C. *3*



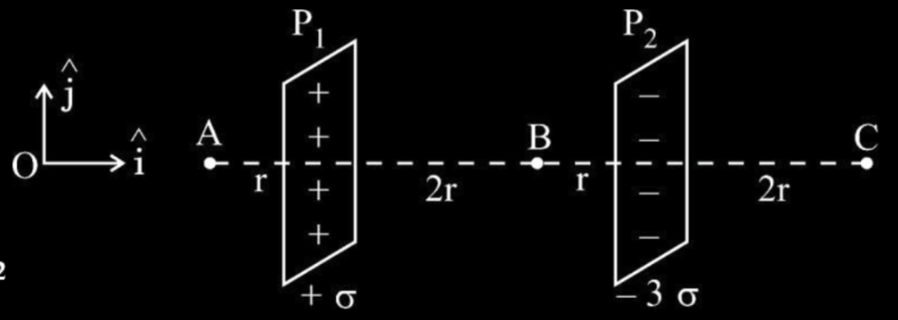
Finding the net electric field (E) at points A,B & C	1+1+1

Electric field at A($\vec{E}_A$)

$$\vec{E}_A = \vec{E}_1 + \vec{E}_2$$ ½

$$= \frac{\sigma}{2\varepsilon_0}(-\hat{i}) + \frac{3\sigma}{2\varepsilon_0}(\hat{i})$$ ½

$$\vec{E}_A = \frac{\sigma}{\varepsilon_0}(\hat{i})$$



Electric field at B ($\vec{E}_B$)

$$\vec{E}_B = \vec{E}_1 + \vec{E}_2$$ ½

$$= \frac{\sigma}{2\varepsilon_0}\hat{i} + \frac{3\sigma}{2\varepsilon_0}\hat{i}$$

$$= \frac{2\sigma}{\varepsilon_0}\hat{i}$$ ½

Electric field at C ($\vec{E}_C$)

$$\vec{E}_c = \vec{E}_1 + \vec{E}_2$$ ½

$$= \frac{\sigma}{2\varepsilon_0}\hat{i} + \frac{3\sigma}{2\varepsilon_0}(-\hat{i})$$ ½

$$= \frac{\sigma}{\varepsilon_0}(-\hat{i})$$

(i) Two identical electric dipoles, each consisting of charges $-q$ and $+q$ separated by distance d, are arranged in x-y plane such that their negative charges lie at the origin O and positive charges lie at points (d, 0) and (0, d) respectively. The net dipole moment of the system is : *1*

(A) $-qd(\hat{i}+\hat{j})$

(B) $qd(\hat{i}+\hat{j})$

(C) $qd(\hat{i}-\hat{j})$

(D) $qd(\hat{j}-\hat{i})$

(i) Two identical electric dipoles, each consisting of charges $-q$ and $+q$ separated by distance d, are arranged in x-y plane such that their negative charges lie at the origin O and positive charges lie at points (d, 0) and (0, d) respectively. The net dipole moment of the system is : *1*

(A) $-qd(\hat{i}+\hat{j})$

(B) $qd(\hat{i}+\hat{j})$

(C) $qd(\hat{i}-\hat{j})$

(D) $qd(\hat{j}-\hat{i})$

(B) $qd(\hat{i}+\hat{j})$

(ii) E_1 and E_2 are magnitudes of electric field due to a dipole, consisting of charges $-q$ and $+q$ separated by distance $2a$, at points r ($\gg a$) (1) on its axis, and (2) on equatorial plane, respectively. Then $\left(\frac{E_1}{E_2}\right)$ is : *1*

(A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) 2 (D) 4

(ii) E_1 and E_2 are magnitudes of electric field due to a dipole, consisting of charges $-q$ and $+q$ separated by distance 2a, at points r (>> a) (1) on its axis, and (2) on equatorial plane, respectively. Then $\left(\frac{E_1}{E_2}\right)$ is : *1*

(A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) 2 (D) 4

(C) 2

(iii) An electric dipole of dipole moment 5.0×10^{-8} Cm is placed in a region where an electric field of magnitude 1.0×10^{3} N/C acts at a given instant. At that instant the electric field $\vec{E}$ is inclined at an angle of 30° to dipole moment $\vec{P}$. The magnitude of torque acting on the dipole, at that instant is : *1*

(A) 2.5×10^{-5} Nm (B) 5.0×10^{-5} Nm

(C) 1.0×10^{-4} Nm (D) 2.0×10^{-6} Nm

(iii) An electric dipole of dipole moment 5.0×10^{-8} Cm is placed in a region where an electric field of magnitude 1.0×10^{3} N/C acts at a given instant. At that instant the electric field $\vec{E}$ is inclined at an angle of 30° to dipole moment $\vec{P}$. The magnitude of torque acting on the dipole, at that instant is : *1*

(A) 2.5×10^{-5} Nm (B) 5.0×10^{-5} Nm

(C) 1.0×10^{-4} Nm (D) 2.0×10^{-6} Nm

(A) 2.5×10^{-5} Nm

(a) (i) Define electric field. Write its SI units. Derive an expression for electric field due to a point charge at a point. Why must the test charge be as small as possible ?

(ii) An electron and an α-particle, both are kept in the same electric field $\vec{E} = E_1 \hat{i} + E_2 \hat{j}$. Find the force acting on them and their acceleration. *5*

(i)Defining electric field	1
SI units	½
Deriving expression for electric field	1
Reason for test charge to be small	½
(ii) Finding the force on electron & α-particle	½+½
Acceleration of electron & α-particle	½+½

(i) Electric field due to a change Q at a point in space may be defined as the force that a unit positive charge would experience if placed at that point.
S I unit is N/C 1

Alternatively-
SI unit is V/m ½

The force between a source charge Q and test charge q will be

$$\vec{F} = \frac{Qq}{4\pi\varepsilon_0 r^2}\hat{r}$$

The electric field due to the source charge

$$\vec{E} = \lim_{q\to 0}\left(\frac{\vec{F}}{q}\right)$$

$$\vec{E} = \frac{Q}{4\pi\varepsilon_0 r^2}\hat{r}$$ 1

- The test charge should be as small as possible so that the source charge remains at its original location. ½

(ii) Electric field $(\vec{E}) = E_1\hat{i} + E_2\hat{j}$ (Given)

Force experienced by the electron $\vec{F}_e = -e\vec{E} = -e\left[E_1\hat{i} + E_2\hat{j}\right]$ ½

Acceleration of the electron $\vec{a}_e = -\frac{\vec{F}_e}{m_e} = -\frac{e}{m_e}\left[E_1\hat{i} + E_2\hat{j}\right]$ ½

Force experienced by an alpha particle
$\vec{F}_\alpha = 2e\vec{E} = 2e\left[E_1\hat{i} + E_2\hat{j}\right]$ ½

Acceleration of the alpha-particle
$\vec{a}_\alpha = \frac{2e}{m_\alpha}\left[E_1\hat{i} + E_2\hat{j}\right]$ ½

(ii) An electric dipole of dipole moment $\vec{p} = (0{\cdot}8\,\hat{i} + 0{\cdot}6\,\hat{j})\,10^{-29}$ Cm is placed in an electric field $\vec{E} = 1{\cdot}0 \times 10^{7}\,\hat{k}\,\frac{V}{m}$. Calculate the magnitude of the torque acting on it and the angle it makes with the x-axis, at this instant.

(ii) $\vec{\tau} = \vec{p} \times \vec{E}$ ½

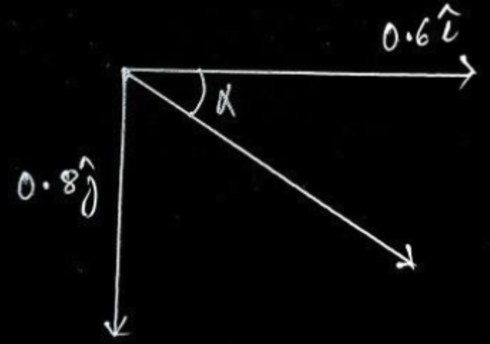
$$= (0.8\hat{i} + 0.6\hat{j}) \times 10^{-29} \times (1 \times 10^{7})\hat{k}$$

$$= \left[0.8\left(-\hat{j}\right) + 0.6\hat{i}\right] \times 10^{-22}$$ ½

$$\tau = \left[\sqrt{(0.8)^2 + (0.6)^2}\right] \times 10^{-22}$$

$$= 10^{-22}\ Nm$$ ½

$$\tan\alpha = \frac{|0.8|}{0.6}$$



$$\alpha = \tan^{-1}\left(\frac{4}{3}\right)$$ ½

$$\alpha = 53^0$$

23. An electric field $\vec{E}$ given by :

$$\vec{E} = 100\,\hat{i}\,\frac{N}{C} \qquad \text{for } x > 0$$

$$= -\,100\,\hat{i}\,\frac{N}{C} \qquad \text{for } x < 0$$

exists in a region. A right circular cylinder of length 10 cm and radius 2 cm, is placed in the region such that its axis coincides with x-axis and its two faces are at x = – 5 cm and x = 5 cm. Calculate :

(a) the net outward flux through the cylinder, and

(b) the net charge inside the cylinder. *3*

Calculating	
(a) Net outward flux through the cylinder	2
(b) Net charge inside the cylinder	1

$$\phi_L = \vec{E} . \overrightarrow{\Delta S}$$

$$= 100\left(-\hat{i}\right) . \Delta S\left(-\hat{i}\right) \qquad \tfrac{1}{2}$$

$$= 100 \times \pi\left(2 \times 10^{-2}\right)^2$$

$$= 4\pi \times 10^{-2} \text{ Nm}^2/\text{C} \qquad \tfrac{1}{2}$$

$$\phi_R = 100\left(\hat{i}\right) . \Delta S\left(\hat{i}\right)$$

$$= 100 \times \pi\left(2 \times 10^{-2}\right)^2$$

$$= 4\pi \times 10^{-2} \text{ Nm}^2/\text{C}$$

$$\phi_{\text{total}} = \phi_{\text{L}} + \phi_{\text{R}} \qquad \tfrac{1}{2}$$

$$= 8\pi \times 10^{-2} \text{ Nm}^2/\text{C} \qquad \tfrac{1}{2}$$

$$= 25.12 \times 10^{-2} \text{ Nm}^2/\text{C} \qquad \tfrac{1}{2}$$

Charge $q = \varepsilon_0 \phi_{total}$

$$= 25.12 \times 10^{-2} \times 8.85 \times 10^{-12} \qquad \tfrac{1}{2}$$

$$= 0.22 \times 10^{-11} \text{ C}$$

(b) (i) Obtain an expression for the electric field $\vec{E}$ due to a dipole of dipole moment $\vec{p}$ at a point on its equatorial plane and specify its direction. Hence, find the value of electric field :

(I) at the centre of the dipole (r = 0), and

(II) at a point r >> a,

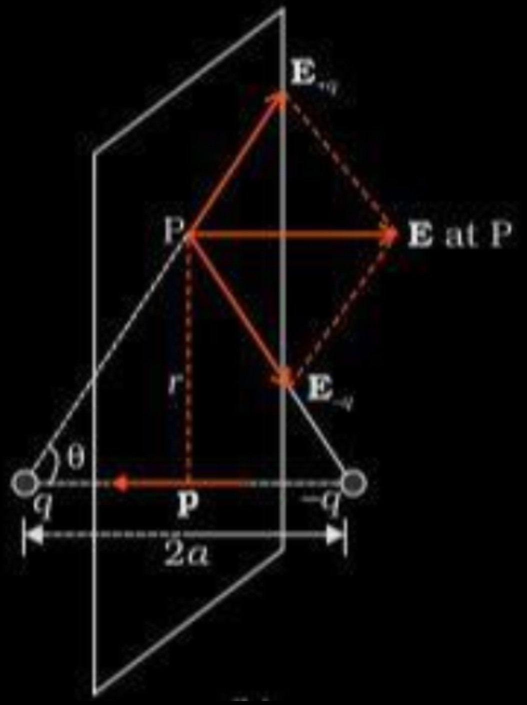
where 2a is the length of the dipole.

(i)	Obtaining expression for electric field due to a dipole on its equatorial plane	2
	Finding electric field:	
	(I) At centre of the dipole	½
	(II) At a point r >> a	½
(ii)	Calculating net electric flux through cube	2

The magnitudes of the electric field due to two charges +q and -q are

$$E_{+q} = \frac{q}{4\pi\varepsilon_0}\frac{1}{\left(r^2+a^2\right)}$$

$$E_{-q} = \frac{q}{4\pi\varepsilon_0}\frac{1}{\left(r^2+a^2\right)}$$ ½



½

The total electric field

$$\vec{E} = -\left(E_{+q} + E_{-q}\right)\cos\theta\,\hat{p}$$

$$\vec{E} = -\frac{\vec{p}}{4\pi\varepsilon_0\left(r^2+a^2\right)^{3/2}}$$ ½

Direction of electric field is opposite to dipole moment ($\vec{p}$) ½

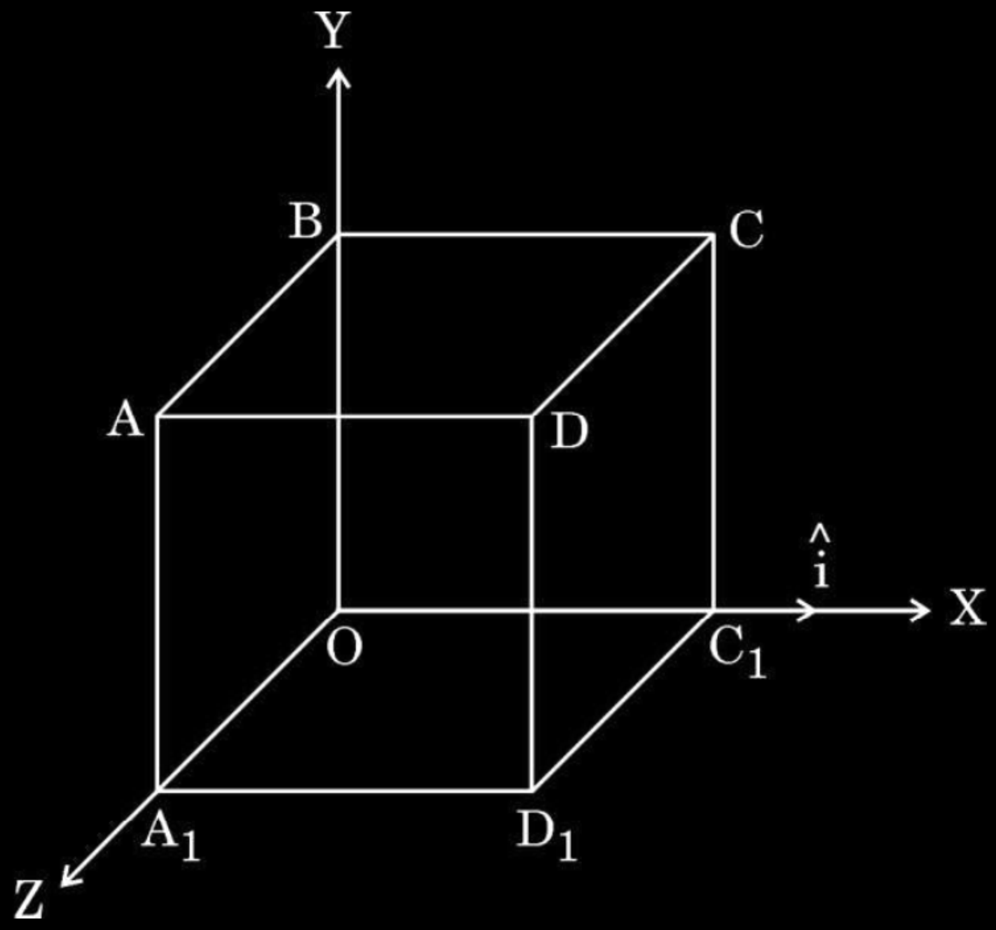
(I) At centre of dipole, r = 0

$$\vec{E} = -\frac{-\vec{p}}{4\pi\varepsilon_0 a^3}$$ ½

(II) At a point r>>a

$$\vec{E} = -\frac{-\vec{p}}{4\pi\varepsilon_0 r^3}$$ ½

(ii) An electric field $\vec{E} = (10x + 5)\hat{i}$ N/C exists in a region in which a cube of side L is kept as shown in the figure. Here x and L are in metres. Calculate the net flux through the cube.



(ii) $\vec{E} = (10x+5)\hat{i}$ N/C

$$\phi_L = \int \vec{E}.\overrightarrow{ds}$$

$$= -E_L(L^2)$$

$$= -5L^2 \qquad \frac{1}{2}$$

$$\phi_R = E_R(L^2)$$

$$= (10L+5)L^2 \qquad \frac{1}{2}$$

$$\phi_{net} = \phi_L + \phi_R$$

$\frac{1}{2}$

$$= -5L^2 + (10L+5)L^2$$

$$= 10L^3 \ \text{Nm}^2/\text{C} \qquad \frac{1}{2}$$

3. A particle of mass m and charge q moving with velocity $v\hat{i}$ is subjected to a uniform electric field $E\hat{j}$. The particle will initially have a tendency to move in a circle of radius :

(A) $\left(\frac{mv}{qE}\right)$ in x-y plane

(B) $\left(\frac{mv^2}{qE^2}\right)$ in x-z plane

(C) $\left(\frac{mv^2}{qE}\right)$ in x-y plane

(D) $\left(\frac{mv}{qE^2}\right)$ in y-z plane

3. A particle of mass m and charge q moving with velocity $v\hat{i}$ is subjected to a uniform electric field $E\hat{j}$. The particle will initially have a tendency to move in a circle of radius :

(A) $\left(\dfrac{mv}{qE}\right)$ in x-y plane

(B) $\left(\dfrac{mv^2}{qE^2}\right)$ in x-z plane

(C) $\left(\dfrac{mv^2}{qE}\right)$ in x-y plane

(D) $\left(\dfrac{mv}{qE^2}\right)$ in y-z plane

(C) $\left[\dfrac{mv^2}{qE}\right]$ *in X – Y plane*

(B) The coulomb force between two point charges is 1.5×10^{-4} N. How will this force be affected in the following situations :

(i) the distance between the charges is doubled.

(ii) the magnitude of each charge is doubled but the distance between the charges remains the same.

(B) The coulomb force between two point charges is 1.5×10^{-4} N. How will this force be affected in the following situations :

(i) the distance between the charges is doubled.

(ii) the magnitude of each charge is doubled but the distance between the charges remains the same.

(b)	(i) Force becomes one forth	1
	(ii) Force becomes four times	1

1. A charge of $-1\ \mu C$ on a body represents 1

(A) loss of 6.25×10^{12} electrons by the body.

(B) gain of 6.25×10^{12} electrons by the body.

(C) gain of 1.6×10^{13} electrons by the body.

(D) loss of 1.6×10^{13} electrons by the body.

1. A charge of $-1\ \mu C$ on a body represents **1**

(A) loss of 6.25×10^{12} electrons by the body.

(B) gain of 6.25×10^{12} electrons by the body.

(C) gain of 1.6×10^{13} electrons by the body.

(D) loss of 1.6×10^{13} electrons by the body.

(B) gain of 6.25×10^{12} electrons by the body

(ii) Two point charges q_1 (= 16 μC) and q_2(= 1 μC) are placed at points $\vec{r}_1 = (3\text{ m})\hat{i}$ and $\vec{r}_2 = (4\text{ m})\hat{j}$. Find the net electric field $\vec{E}$ at point $\vec{r} = (3\text{ m})\hat{i} + (4\text{ m})\hat{j}$.

(ii)

$$\vec{E} = \frac{Kq}{r^2}\,\hat{r}$$ ½

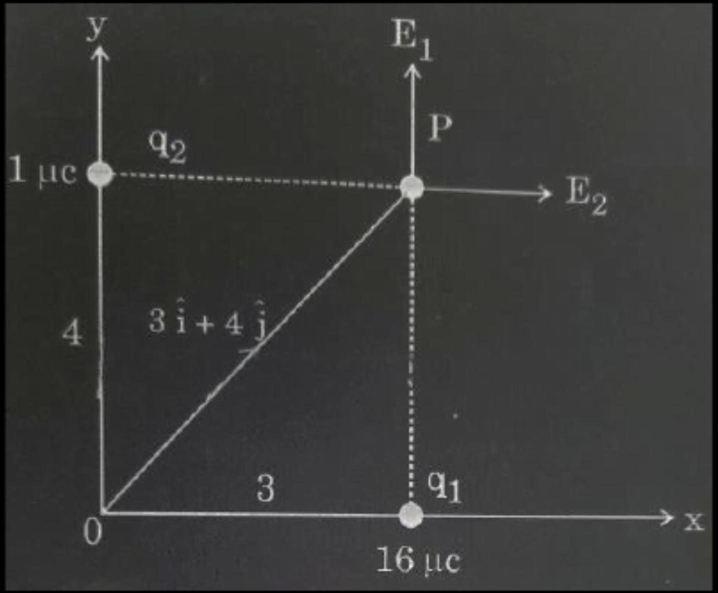
$$\vec{E}_1 = \frac{9\times10^9\times16\times10^{-6}}{(4)^2}\,\hat{j}$$ ½

$$= 9\times10^3\ \hat{j}$$

$$\vec{E}_2 = \frac{9\times10^9\times1\times10^{-6}}{(3)^2}\,\hat{i}$$ ½

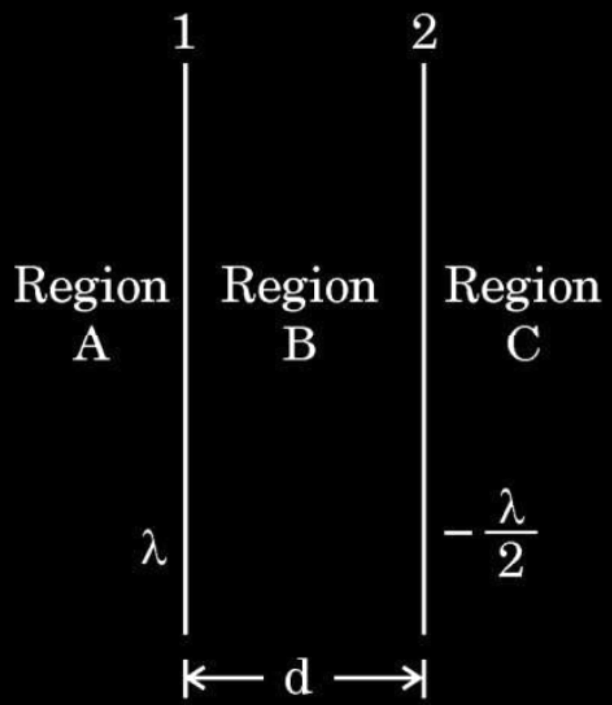
$$= 10^3\ \hat{i}$$

$$\vec{E}_{net} = (\hat{i} + 9\hat{j})\,10^3\ N/C$$ ½

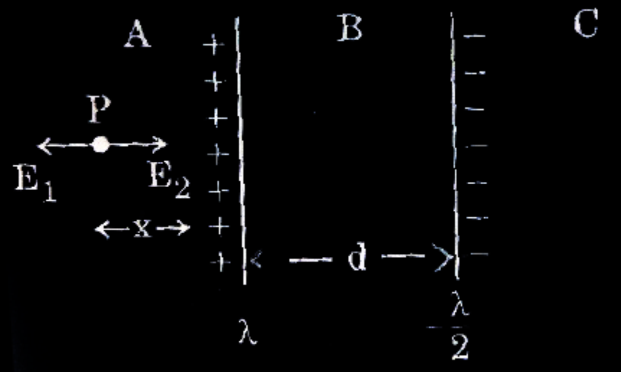


NOTE: Award full credit of this part if a student finds magnitude and direction separately.

(b) Two infinitely long straight wires '1' and '2' are placed d distance apart, parallel to each other, as shown in the figure. They are uniformly charged having charge densities λ and $-\frac{\lambda}{2}$ respectively. Locate the position of the point from wire '1' at which the net electric field is zero and identify the region in which it lies. *3*



Location of point at which net electric field is zero	2½
Identification of Region	½



½

Electric field due to wire 1 and wire 2 at point P

$$E_1 = \frac{\lambda}{2\pi\varepsilon_0 x}$$ ½

$$E_2 = \frac{\lambda/2}{2\pi\varepsilon_0(x+d)}$$ ½

At P, Net electric field is zero ½

$$E_1 = E_2$$

$$\frac{\lambda}{2\pi\varepsilon_0 x} = \frac{\lambda}{2\times 2\pi\varepsilon_0(x+d)}$$

$$x = -2d$$ ½

Negative sign indicates that point lies in the region C. ½

At a distance 2d from wire 1 electric field is zero.

(**Note** : Award full credit if a student finds the position by taking point in region C directly)

(b) (i) Show that Gauss's theorem is consistent with Coulomb's law. Using it, derive an expression for the electric field due to a uniformly charged thin spherical shell of radius r at a point at a distance y from the centre of the shell such that (I) $y > r$, and (II) $y < r$.

(i)

- Showing consistency of Gauss's theorem with Coulomb's law 1
- Derivation for electric field due to uniformly charged thin spherical shell at (I) $y > r$ (II) $y < r$ 2

(i)

- Gauss's theorem is based on the inverse square dependence on distance contained in the coulomb's law. 1

  Alternatively-

  According to Gauss's theorem

$$\oint \vec{E}.d\vec{s} = \frac{q}{\varepsilon_o}$$

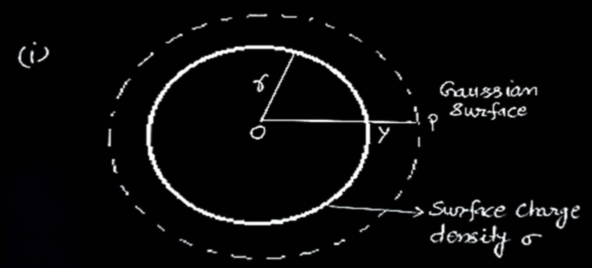
$$E = \frac{1}{4\pi\varepsilon_o}\frac{q}{r^2}$$ ½

According to Coulomb's law, force on charge q_o in this field

$$F = \frac{1}{4\pi\varepsilon_o}\frac{qq_o}{r^2}$$

Therefore, Gauss's law is consistent with Coulomb's law ½

- (I) For y > r



Electric flux through Gaussian surface $E\times 4\pi y^2$

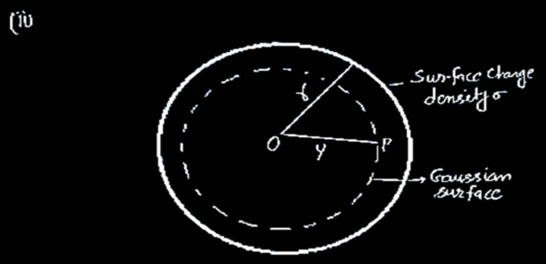
The charge enclosed by the surface $\sigma\times 4\pi r^2$

Using Gauss theorem

$$E(4\pi y^2) = \frac{\sigma 4\pi r^2}{\varepsilon_\circ}$$ ½

$$\vec{E} = \frac{q}{4\pi\varepsilon_0 y^2}\hat{r}$$ ½

(II) For y < r



The charge enclosed by Gaussian surface = 0

Using Gauss theorem

Electric flux $= E(4\pi y^2) = 0$ ½

i.e. $E = 0 \quad (y < r)$ ½

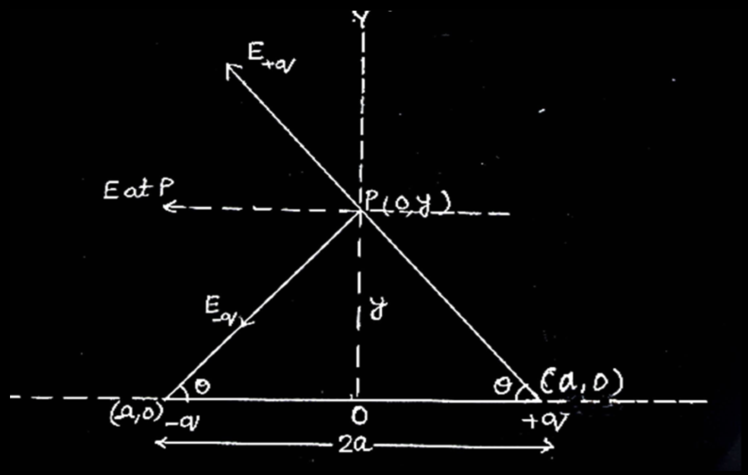
32. (a) (i) Two point charges + q and – q are held at (a, 0) and (– a, 0) in x-y plane. Obtain an expression for the net electric field due to the charges at a point (0, y). Hence, find electric field at a far off point (y >> a).

(i) Finding electric field at a far off point (y>>a)	3
(ii) Calculation of work done in shifting the charges	2

Magnitude of electric field due to the two charges +q and -q are given by

$$E_{+q} = \frac{q}{4\pi\varepsilon_o} \frac{1}{y^2+a^2}$$ ½

$$E_{-q} = \frac{q}{4\pi\varepsilon_o} \frac{1}{y^2+a^2}$$ ½



½

Components normal to the dipole axis cancel out.

The components along the dipole axis add up.

The total electric field is opposite to the dipole moment will be given by-

$$\vec{E} = -(E_{+q}+E_{-q}) \cos\theta\, \hat{p}$$ ½

$$= -\frac{2qa}{4\pi\varepsilon_o (y^2+a^2)^{3/2}} \hat{p}$$ ($\hat{p}$ is a unit vector along dipole moment) ½

At large distance (y>>a)

$$\vec{E} = \frac{-2qa}{4\pi\varepsilon_o y^3} \hat{p}$$ ½

1. Two point charges Q and – q are held 'r' distance apart in free space. A uniform electric field $\vec{E}$ is applied in the region perpendicular to the line joining the two charges. Which one of the following angles will the direction of the net force acting on charge – q make with the line joining Q and – q ?

(A) $\tan^{-1} \dfrac{4\pi\varepsilon_0 E r^2}{Q}$

(B) $\cot^{-1} \dfrac{4\pi\varepsilon_0 E r^2}{Q}$

(C) $\tan^{-1} \dfrac{QE}{4\pi\varepsilon_0 r^2}$

(D) $\cot^{-1} \dfrac{QE}{4\pi\varepsilon_0 r^2}$

1. Two point charges Q and – q are held 'r' distance apart in free space. A uniform electric field $\vec{E}$ is applied in the region perpendicular to the line joining the two charges. Which one of the following angles will the direction of the net force acting on charge – q make with the line joining Q and – q ?

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(B) $\cot^{-1} \dfrac{4\pi\varepsilon_0 E r^2}{Q}$

(C) $\tan^{-1} \dfrac{QE}{4\pi\varepsilon_0 r^2}$

(D) $\cot^{-1} \dfrac{QE}{4\pi\varepsilon_0 r^2}$

(A) $\tan^{-1} \dfrac{4\pi\varepsilon_o E r^2}{Q}$

1. An electric dipole of dipole moment $1 \cdot 0 \times 10^{-12}$ Cm lies along x-axis. An electric field of magnitude $2 \cdot 0 \times 10^{4}$ NC^{-1} is switched on at an instant in the region. The unit vector along the electric field is $\frac{\sqrt{3}}{2}\hat{i} + \frac{1}{2}\hat{j}$. The magnitude of the torque acting on the dipole at that instant is :

(A) $0 \cdot 5 \times 10^{-6}$ Nm (B) $1 \cdot 0 \times 10^{-8}$ Nm

(C) $2 \cdot 0 \times 10^{-8}$ Nm (D) $4 \cdot 0 \times 10^{-8}$ Nm

1. An electric dipole of dipole moment $1{\cdot}0 \times 10^{-12}$ Cm lies along x-axis. An electric field of magnitude $2{\cdot}0 \times 10^{4}$ NC^{-1} is switched on at an instant in the region. The unit vector along the electric field is $\frac{\sqrt{3}}{2}\hat{i} + \frac{1}{2}\hat{j}$. The magnitude of the torque acting on the dipole at that instant is :

(A) $0{\cdot}5 \times 10^{-6}$ Nm (B) $1{\cdot}0 \times 10^{-8}$ Nm

(C) $2{\cdot}0 \times 10^{-8}$ Nm (D) $4{\cdot}0 \times 10^{-8}$ Nm

(B) 1.0×10^{-8}Nm

1. A body acquires charge $8{\cdot}0 \times 10^{-12}$ C. The mass of the body :

(A) increases by $4{\cdot}5 \times 10^{-7}$ kg

(B) decreases by $1{\cdot}0 \times 10^{-6}$ kg

(C) decreases by $4{\cdot}55 \times 10^{-23}$ kg

(D) increases by $9{\cdot}1 \times 10^{-23}$ kg

1. A body acquires charge $8 \cdot 0 \times 10^{-12}$ C. The mass of the body :

(A) increases by $4 \cdot 5 \times 10^{-7}$ kg

(B) decreases by $1 \cdot 0 \times 10^{-6}$ kg

(C) decreases by $4 \cdot 55 \times 10^{-23}$ kg

(D) increases by $9 \cdot 1 \times 10^{-23}$ kg

(C) decreases by 4.55×10^{-23} kg

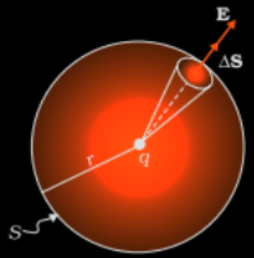
(b) (i) What is difference between an open surface and a closed surface ? Draw elementary surface vector $d\vec{S}$ for a spherical surface S.

(ii) Define electric flux through a surface. Give the significance of a Gaussian surface. A charge outside a Gaussian surface does not contribute to total electric flux through the surface. Why ?

(iii) A small spherical shell S_1 has point charges $q_1 = -3\ \mu C$, $q_2 = -2\ \mu C$ and $q_3 = 9\ \mu C$ inside it. This shell is enclosed by another big spherical shell S_2. A point charge Q is placed in between the two surfaces S_1 and S_2. If the electric flux through the surface S_2 is four times the flux through surface S_1, find charge Q.

(i) Difference between an open surface and a closed surface	½
Diagram of elementary surface vector $\vec{ds}$	1
(ii) Definition of electric flux	1
Significance of Gaussian Surface	½
Reason	½
(iii) Finding charge Q	1½

(i) Open Surface – A surface which does not enclose a volume.
Closed Surface – A surface which does enclose a volume. ½



1

(ii) Electric flux is defined as the number of electric field lines crossing an area normally. 1

Alternatively-

$$\phi = \vec{E}.\vec{A}$$

Alternatively-

$$\phi = EA\cos\theta$$

Significance of Gaussian Surface: -

It helps in finding the electric field in a simpler way. ½

Reason: -

Because any electric field line from the charge which enters the surface at one point will exit at another, resulting in a net zero flux. ½

(iii) Total charge enclosed by $S_1 = (-3-2+9)\ \mu C = 4\ \mu C$ ½

Total charge enclosed by $S_2 = Q + 4\ \mu C$

$$\phi_{s_2} = 4\phi_{s_1}$$ ½

$$\frac{Q + 4\mu C}{\varepsilon_0} = 4\left(\frac{4\mu C}{\varepsilon_0}\right)$$

$Q = 12\ \mu C$ ½

1. Consider two identical dipoles D_1 and D_2. Charges $-q$ and q of dipole D_1 are located at (0, 0) and (a, 0) and that of dipole D_2 at (0, a) and (0, 2a) in x-y plane, respectively. The net dipole moment of the system is **1**

(A) $qa(\hat{i} + \hat{j})$

(B) $-qa(\hat{i} + \hat{j})$

(C) $qa(\hat{i} - \hat{j})$

(D) $-qa(\hat{i} - \hat{j})$

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(A) $qa(\hat{i}+\hat{j})$

1. Two identical point charges are placed at the two vertices A and B of an equilateral triangle of side l. The magnitude of the electric field at the third vertex P is E. If a hollow conducting sphere of radius (l/4) is placed at P, the magnitude of the electric field at point P now becomes **1**

(A) >E

(B) E

(C) $\frac{E}{2}$

(D) zero

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(A) >E (B) E

(C) $\frac{E}{2}$ (D) zero

(D) Zero

1. Two charges –q each are placed at the vertices A and B of an equilateral triangle ABC. If M is the mid-point of AB, the net electric field at C will point along 1

(A) CA (B) CB

(C) MC (D) CM

(D) CM

(b) (i) Consider three metal spherical shells A, B and C, each of radius R. Each shell is having a concentric metal ball of radius R/10. The spherical shells A, B and C are given charges +6q, –4q, and 14q respectively. Their inner metal balls are also given charges –2q, +8q and –10q respectively. Compare the magnitude of the electric fields due to shells A, B and C at a distance 3R from their centres.

Total charge for A = Total charge for B = Total charge for C = $+4q$ 1

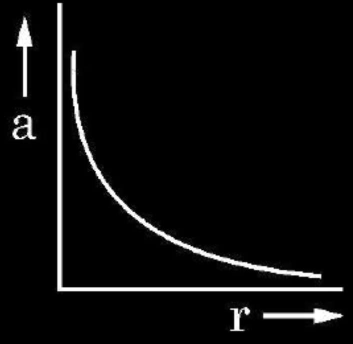
As , $E = \dfrac{kQ}{r^2}$

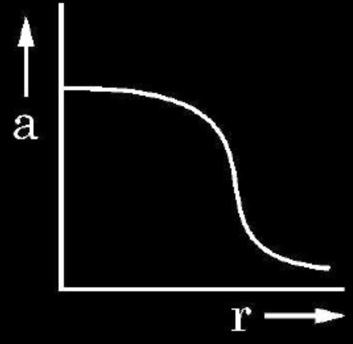
Since $Q = 4q$ and $r = 3R$ ½

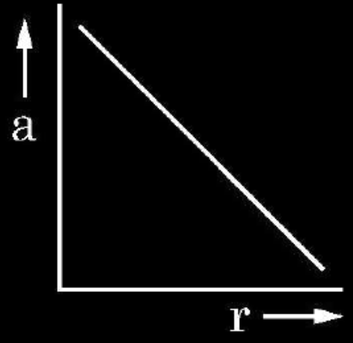
$$E = \frac{k(4q)}{9R^2} = \frac{4kq}{9R^2}$$ ½

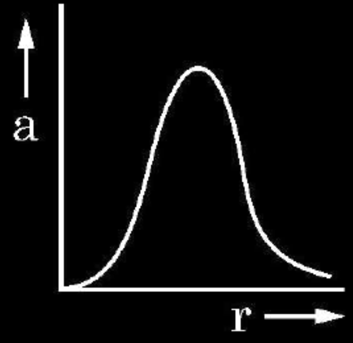
$\therefore\ E_A = E_B = E_c$

1. A charge Q is fixed in position. Another charge q is brought near charge Q and released from rest. Which of the following graphs is the correct representation of the acceleration of the charge q as a function of its distance r from charge Q ? **1**

(A) 

(B) 

(C) 

(D) 

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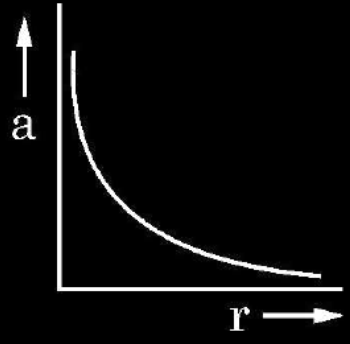
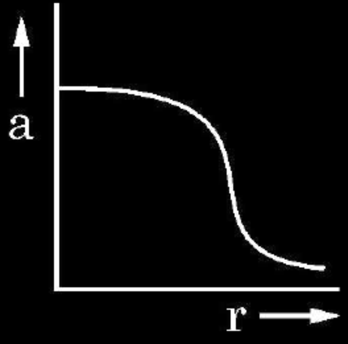
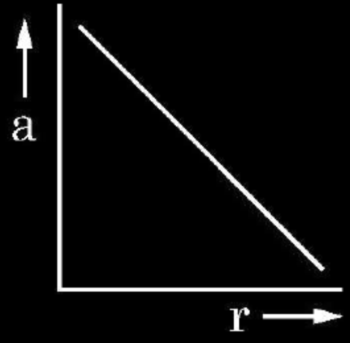
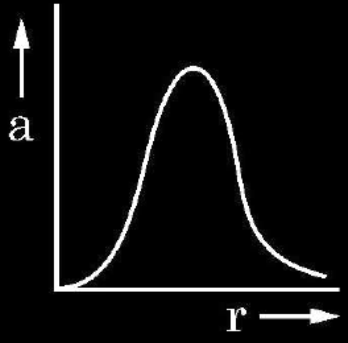
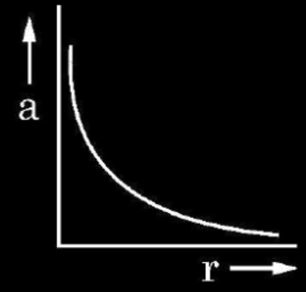
(A) 

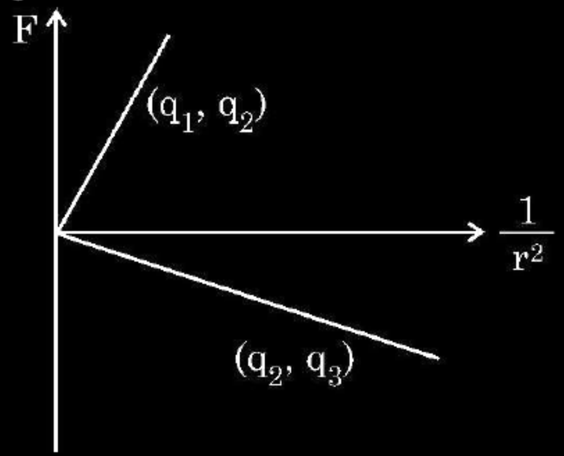
(B) 

(C) 

(D) 

(A) 

1. Figure shows variation of Coulomb force (F) acting between two point charges with $\frac{1}{r^2}$, r being the separation between the two charges (q_1, q_2) and (q_2, q_3). If q_2 is positive and least in magnitude, then the magnitudes of q_1, q_2 and q_3 are such that 1



(A) $q_2 < q_3 < q_1$ (B) $q_3 < q_1 < q_2$

(C) $q_1 < q_2 < q_3$ (D) $q_2 < q_1 < q_3$

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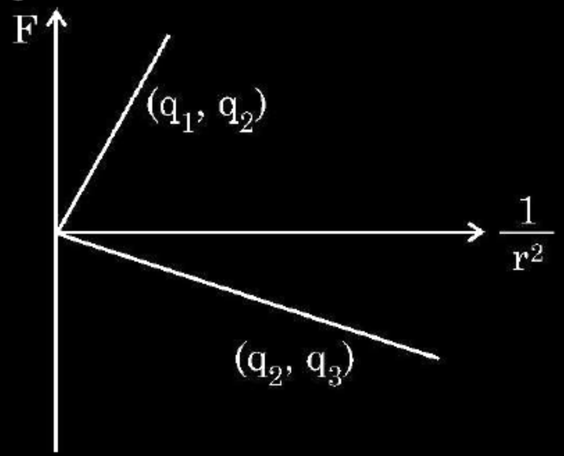

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(C) $q_1 < q_2 < q_3$ (D) $q_2 < q_1 < q_3$

(A) $q_2 < q_3 < q_1$

7. The electric flux through a closed Gaussian surface depends upon

(a) Net charge enclosed and permittivity of the medium

(b) Net charge enclosed, permittivity of the medium and the size of the Gaussian surface

(c) Net charge enclosed only

(d) Permittivity of the medium only 1

7. The electric flux through a closed Gaussian surface depends upon

(a) Net charge enclosed and permittivity of the medium

(b) Net charge enclosed, permittivity of the medium and the size of the Gaussian surface

(c) Net charge enclosed only

(d) Permittivity of the medium only **1**

(a) Net Charge enclosed and permittivity of the medium

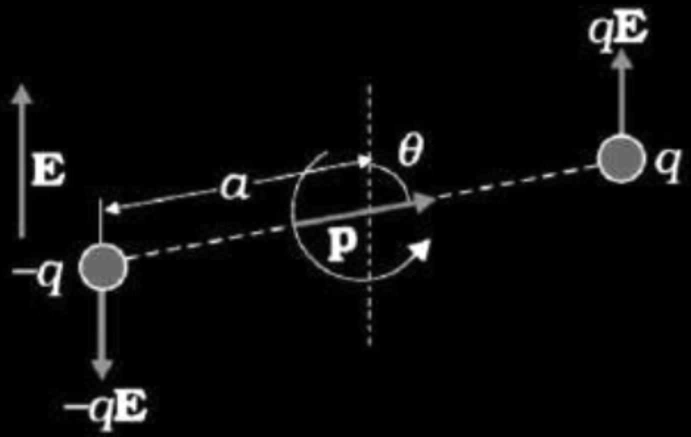
13. A point charge is placed at the centre of a hollow conducting sphere of internal radius ‘r’ and outer radius ‘2r’. The ratio of the surface charge density of the inner surface to that of the outer surface will be ________. **1**

13. A point charge is placed at the centre of a hollow conducting sphere of internal radius 'r' and outer radius '2r'. The ratio of the surface charge density of the inner surface to that of the outer surface will be ________. **1**

$\boxed{4:1}$

21. Derive the expression for the torque acting on an electric dipole, when it is held in a uniform electric field. Identify the orientation of the dipole in the electric field, in which it attains a stable equilibrium. **2**

Derivation of the expression for the torque	1½
Identification of the orientation of stable equilibrium	½



½

Force on q is q**E** and a force on –q is –q**E**.

Hence torque

$\tau = qE \times 2a \sin\theta$ ½

$\tau = PE \sin\theta$

½

$\vec{\tau} = \vec{P} \times \vec{E}$

For stable equilibrium $\theta = 0^0$ ½

(a) State Gauss's law on electrostatics and derive an expression for the electric field due to a long straight thin uniformly charged wire (linear charge density λ) at a point lying at a distance r from the wire.

(a)	Statement of Gauss's law	1
	Derivation of the expression of the electric field	2½

(a) Gauss law: Electric flux through of a closed surface is $\frac{1}{\varepsilon_0}$ times the charge enclosed by the surface.

Alternatively

$$\phi_E = \frac{q}{\varepsilon_0}$$ 1

Let the charge be uniformly distributed on a wire

$$\phi = \oint d\phi = \int_{s_1} \vec{E} \cdot d\vec{s}_1 + \int_{s_2} \vec{E} \cdot d\vec{s}_2 + \int_{s_3} \vec{E} \cdot d\vec{s}_3$$ ½

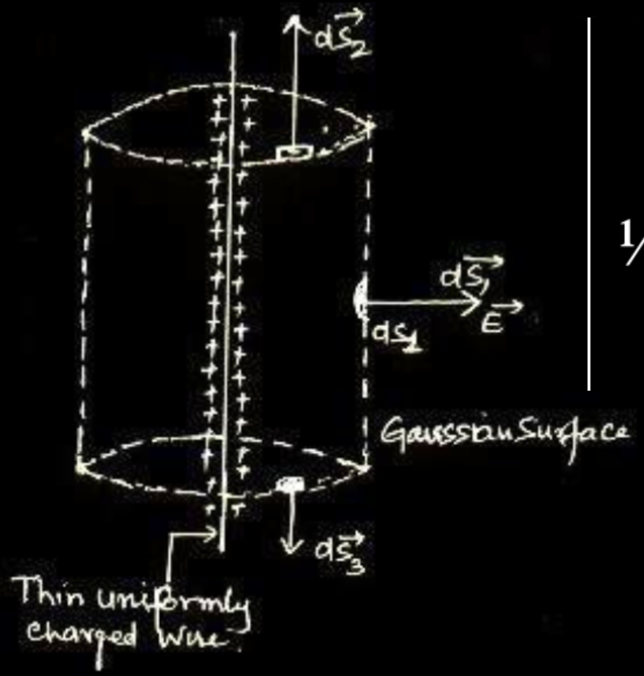
$$= \int_{s_1} Eds_1 \cos 0^0 + \int_{s_2} Eds_2 \cos 90^0 + \int_{s_3} Eds_3 \cos 90^0$$ ½

$$= E \int_{s_1} ds_1 = E.2\pi rl$$

byGauss'slaw

$$\frac{q}{\varepsilon_0} = E \cdot 2\pi rl$$ ½

$$E = \frac{q}{2\pi\varepsilon_0 rl} = \frac{1}{2\pi\varepsilon_0}\frac{\lambda}{r}$$ ½



½

11. Torque acting on an electric dipole placed in an electric field is maximum when the angle between the electric field and the dipole moment is _______. 1

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90^0 or $\frac{\pi}{2}$

28. A hollow conducting sphere of inner radius r_1 and outer radius r_2 has a charge Q on its surface. A point charge $-q$ is also placed at the centre of the sphere.

(a) What is the surface charge density on the (i) inner and (ii) outer surface of the sphere ?

(b) Use Gauss' law of electrostatics to obtain the expression for the electric field at a point lying outside the sphere. **3**

(a) Giving the value of surface charge density of (i) Inner surface (ii) Outer Surface	½ + ½
(b) Deriving expression for electric field	2

(a) Surface charge density on the inner surface $= \frac{q}{4\pi r_1^2}$ ½

On the outer surface $= \frac{Q-q}{4\pi r_2^2}$ ½

(b) For a spherical Gaussian surface $x > r_2$

$$\oint \vec{E}\,\overrightarrow{ds} = \frac{Q-q}{\epsilon_0}$$ 1

$$E \times 4\pi x^2 = \frac{Q-q}{\epsilon_0}$$ ½

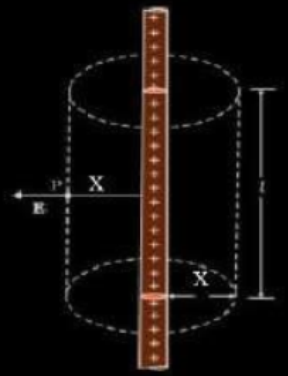
$$E = \frac{1}{4\pi\epsilon_0}\frac{Q-q}{x^2}$$ ½

(a) An infinitely long thin straight wire has a uniform linear charge density λ. Obtain the expression for the electric field (E) at a point lying at a distance x from the wire, using Gauss' law.

(b) Show graphically the variation of this electric field E as a function of distance x from the wire. **3**

(a)	Derivation for electric field due to a uniformly charged straight wire	2
(b)	Graph showing variation of electric field E vs distance x	1

(a)



½

$$\oint \vec{E}.\vec{dS} = \frac{q}{\epsilon_0}$$

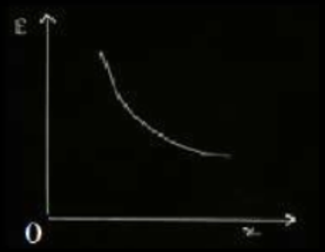
$$\int \vec{E}\,.\,\vec{dS_1} + \int \vec{E}\,.\,\vec{dS_2} + \int \vec{E}\,.\,\vec{dS_3} = \frac{\lambda l}{\epsilon_0}$$

$$E\; dS_1 \cos 90^0 + \; EdS_2\; \cos 90^0 \; + \; EdS_3 \cos 0^0 = \; \frac{\lambda l}{\epsilon_0}$$ ½

$$0\; + 0\; + E\; \times 2\pi x l = \frac{\lambda l}{\epsilon_0}$$ ½

$$E \; = \; \frac{\lambda}{2\pi\epsilon_0 x}$$ ½

(b)



1

2. The electric flux emerging out from 1 C charge is

(A) $\frac{1}{\varepsilon_0}$

(B) 4π

(C) $\frac{4\pi}{\varepsilon_0}$

(D) ε_0

2. The electric flux emerging out from 1 C charge is

(A) $\dfrac{1}{\varepsilon_0}$

(B) 4π

(C) $\dfrac{4\pi}{\varepsilon_0}$

(D) ε_0

(A)

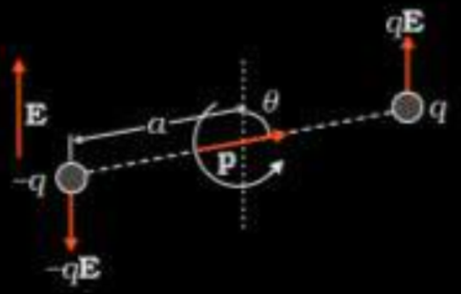
$\dfrac{1}{\epsilon_o}$

35. (a) An electric dipole of dipole moment $\vec{p}$ is placed in a uniform electric field $\vec{E}$ at an angle θ with it. Derive the expression for torque ($\vec{\tau}$) acting on it. Find the orientation of the dipole relative to the electric field for which torque on it is (i) maximum, and (ii) half of maximum.

(b) Two point charges $q_1 = +1\ \mu C$ and $q_2 = +4\ \mu C$ are placed 2 m apart in air. At what distance from q_1 along the line joining the two charges, will the net electric field be zero? *5*

a) Diagram	½ mark
Derivation	1 ½ mark
Orientation for maximum and half of the maximum torque	½ + ½ mark
b) Formula	½ mark
Calculation	1 mark
Result	½ mark

a)



From diagram

Magnitude of Torque= $(qE)(2a\ sin\theta)$

$$= (2qa)(E\ sin\theta)$$

$$= pEsin\theta$$

For direction

$$\vec{\tau} = \vec{p} \times \vec{E}$$

i) for maximum Torque, dipole should be placed perpendicular to the direction of electric field

$$\theta = 90^0 = \frac{\pi}{2}$$

ii) For the torque to be half the maximum,

$$\theta = 30^0 = \frac{\pi}{6}$$

½

½

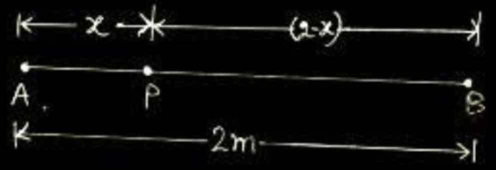
½

½

½

½

(b)



$$E_{PA} = E_{PB} \quad ; \quad E = \frac{kq}{r^2}$$

$$\frac{kq_A}{x^2} = \frac{kq_B}{(2-x)^2}$$ ½

$$\frac{1}{x^2} = \frac{4}{2-x)^2}$$ ½

$$\frac{1}{x} = \frac{2}{2-x}$$ ½

$$x = \frac{2}{3}\ m$$ ½

11. If the electric flux entering and leaving a closed surface in air are ϕ_1 and ϕ_2 respectively, the net electric charge enclosed within the surface is __________ . *1*

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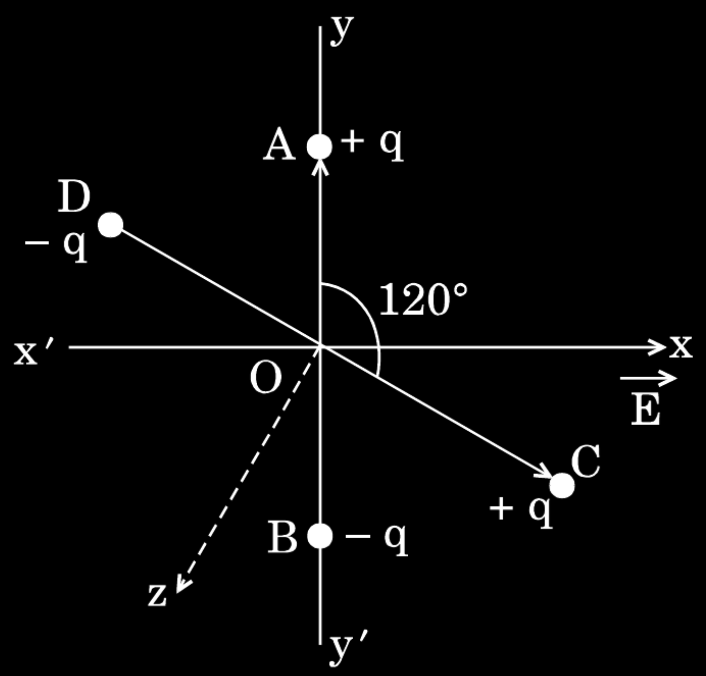
$(\emptyset_2 - \emptyset_1)\varepsilon_0 /\ (\emptyset_1 - \emptyset_2)\varepsilon_0$

28. Two small identical electric dipoles AB and CD, each of dipole moment $\vec{p}$ are kept at an angle of 120° to each other in an external electric field $\vec{E}$ pointing along the x-axis as shown in the figure. Find the

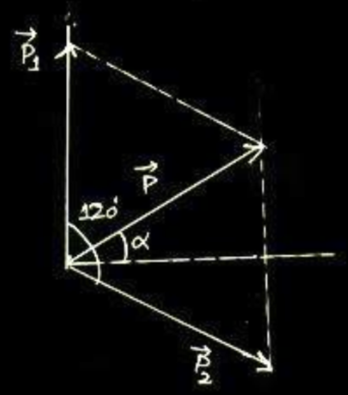
(a) dipole moment of the arrangement, and

(b) magnitude and direction of the net torque acting on it. *3*



(a) Magnitude & direction of net dipole moment	1½
(b) Magnitude & direction of net torque	1½

(a)



½

$$P = (p_1{}^2 + p_2{}^2 + 2\,p_1\,p_2\,\cos 120^0)^{1/2}$$
$$= (2p^2 - p^2)^{1/2}$$
$$= p$$

½

Making 60^0 angle with $\overrightarrow{p_1}$ and $\alpha = 30^0$ (angle with X axis) ½

[$p_1 = p_2 = p$]

[Do not deduct ½ mark if diagram is not drawn but dipole moment and its direction are correctly worked out.

If correct and complete vector diagram is drawn but dipole moment is not worked out then award 1 mark out of 1.5] ½

(b)

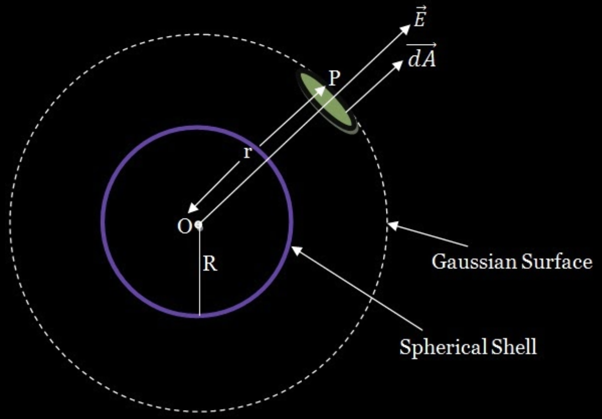
$$\vec{\tau} = \vec{P} \times \vec{E}$$ ½
$$\tau = PE\ \sin 30^0$$
$$= \frac{1}{2}\ pE$$ ½

Direction of $\vec{\tau}$ is into the plane of the paper or along -z direction.

35. (a) Use Gauss's law to show that due to a uniformly charged spherical shell of radius R, the electric field at any point situated outside the shell at a distance r from its centre is equal to the electric field at the same point, when the entire charge on the shell were concentrated at its centre. Also plot the graph showing the variation of electric field with r, for $r \leq R$ and $r \geq R$.

(b) Two point charges of $+1\ \mu C$ and $+4\ \mu C$ are kept 30 cm apart. How far from the $+1\ \mu C$ charge on the line joining the two charges, will the net electric field be zero ? *5*

(a) Expression for electric field outside a charged shell	2
Graph of E vs r	1
b) Location of point where field is zero	2



½

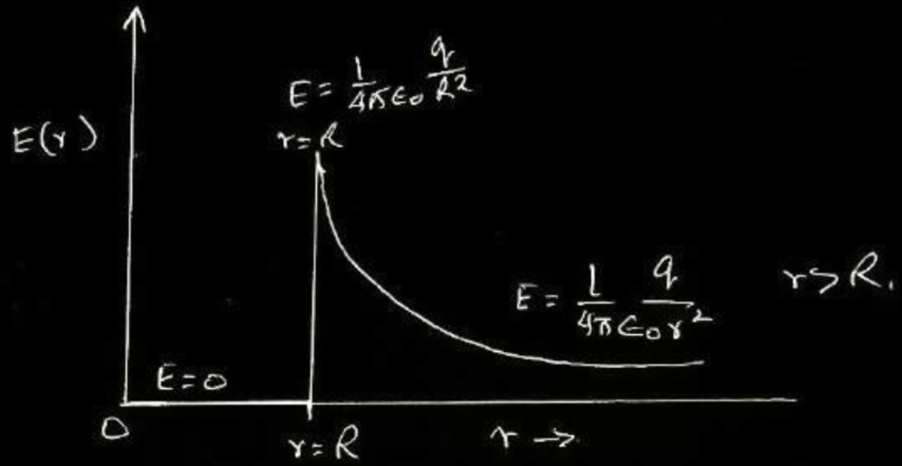
$$\phi = \frac{q}{\epsilon_0}$$ ½

$$E \times 4\pi r^2 = \frac{\sigma(4\pi R^2)}{\epsilon_0}$$ ½

$$E = \frac{1}{4\pi\epsilon_o}\frac{q}{r^2}$$ ½

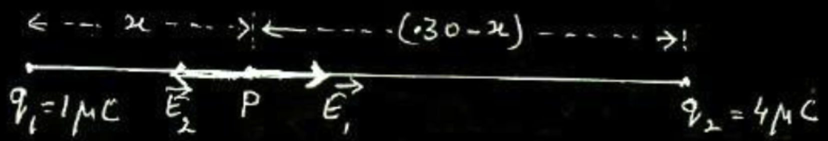
$$[\therefore q = \sigma(4\pi R^2)]$$

which is electric field due to a point charge q at a distance r from it



1

(b)



½

$$E_1 = E_2$$

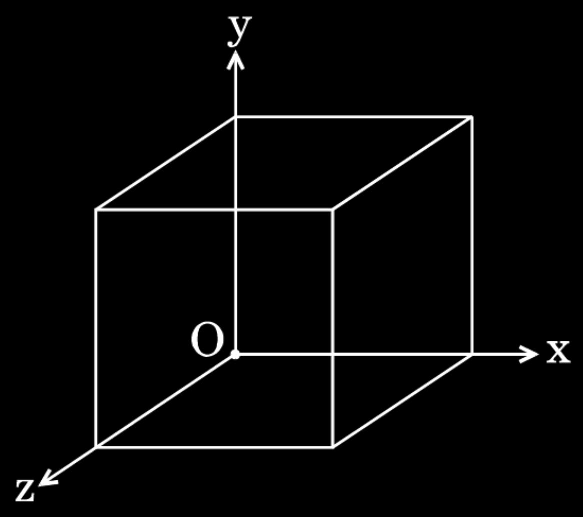
$$\frac{1}{4\pi\epsilon_o}\frac{1 \times 10^{-6}}{x^2} = \frac{1}{4\pi\epsilon_o}\frac{4 \times 10^{-6}}{(0.3 - x)^2}$$ ½

$$(0.3 - x)^2 = 4x^2$$ ½

$$0.3 - x = 2x$$

$$x = 0.1m = 10\ cm\ (to\ the\ right\ of\ q_1)$$ ½

(b) A cube of side 20 cm is kept in a region as shown in the figure. An electric field $\vec{E}$ exists in the region such that the potential at a point is given by V = 10x + 5, where V is in volt and x is in m.



Find the

(i) electric field $\vec{E}$, and

(ii) total electric flux through the cube.

(b)

(i)

$$E = \frac{-dV}{dx} = -\frac{d}{dx}(10x + 5)$$ 1

$$\therefore \vec{E} = -10\hat{i}\ N/C$$ ½

(ii) Electric flux through the cube, ϕ =sum of electric flux through 6 faces

Electric flux through faces perpendicular Y and Z axis = 0 ½

$\because$ E is along x axis

Electric flux through faces perpendicular to x axis

$$= \phi_1 + \phi_2$$ ½

$$= 10 \times (0.2)^2 - 10 \times (0.2)^2$$

$$=0$$ ½

1. If the net electric flux through a closed surface is zero, then we can infer *1*

(A) no net charge is enclosed by the surface.

(B) uniform electric field exists within the surface.

(C) electric potential varies from point to point inside the surface.

(D) charge is present inside the surface.

1. If the net electric flux through a closed surface is zero, then we can infer *1*

(A) no net charge is enclosed by the surface.

(B) uniform electric field exists within the surface.

(C) electric potential varies from point to point inside the surface.

(D) charge is present inside the surface.

(A)

no net charge is enclosed by the surface

35. (a) Using Gauss law, derive expression for electric field due to a spherical shell of uniform charge distribution σ and radius R at a point lying at a distance x from the centre of shell, such that

(i) $0 < x < R$, and

(ii) $x > R$.

(b) An electric field is uniform and acts along + x direction in the region of positive x. It is also uniform with the same magnitude but acts in – x direction in the region of negative x. The value of the field is $E = 200$ N/C for $x > 0$ and $E = -200$ N/C for $x < 0$. A right circular cylinder of length 20 cm and radius 5 cm has its centre at the origin and its axis along the x-axis so that one flat face is at $x = +10$ cm and the other is at $x = -10$ cm.

Find :

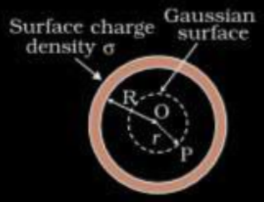
(i) The net outward flux through the cylinder.

(ii) The net charge present inside the cylinder. *5*

(a) (i) Electric Field inside hollow sphere	1½ mark
(ii) Electric Field outside hollow sphere	1½ mark
(b) (i) The net outward flux through cylinder	1 mark
(ii) The net charge present inside the cylinder	1 mark

(a)

(i)



½

According to Gauss's Law

$$\oint \vec{E}.\overrightarrow{dA} = \frac{q_{in}}{\epsilon_0}$$ ½

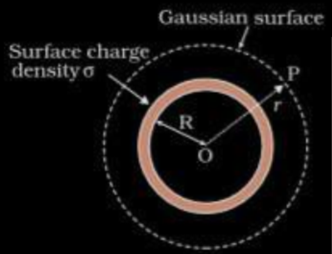
$\because$ *inside hollow sphere*

$q_{in} = 0$

$$\therefore \oint \vec{E}\,.\overrightarrow{dA} = 0$$

E = 0 ½

(ii)



½

$$q = \sigma 4\pi R^2$$

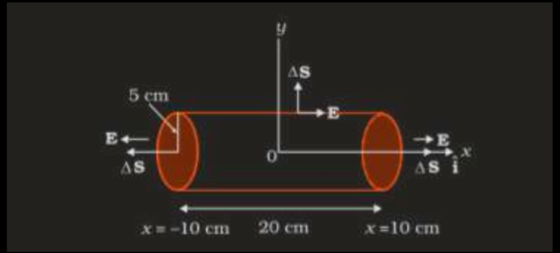
$$\oint \overrightarrow{E.dA} = \frac{q}{\epsilon_0}$$ ½

$$E\oint dA = \frac{q}{\epsilon_0}$$

$$E.4\pi x^2 = \frac{\sigma 4\pi R^2}{\epsilon_0}$$

$$E = \frac{\sigma R^2}{\epsilon_0 x}$$ ½

b)



(i) The net outward flux through cylinder

$$\phi = EA + EA = 2EA \qquad A = \pi r^2$$

$= 2 \times 200 \times 3.14 \times 0.05 \times 0.05$ ½

$$= 3.14\,\frac{N}{C}m^2$$ ½

(ii) The net charge present inside the cylinder

$$q = \epsilon_0 \phi$$ ½

$q = 8.854 \times 3.14 \times 10^{-12}$ ½

$= 2.78 \times 10^{-11}$ C

Derive an expression for the torque acting on an electric dipole of dipole moment $\vec{p}$ placed in a uniform electric field $\vec{E}$. Write the direction along which the torque acts.

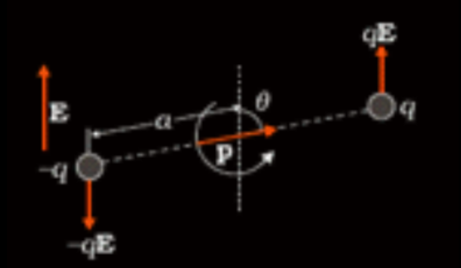
OR

Derive an expression for the electric field at a point on the axis of an electric dipole of dipole moment $\vec{p}$. Also write its expression when the distance r >> the length 'a' of the dipole.

Diagram -	½
Expression for torque -	1
Direction of torque -	½

Expression for field at an axial point - 1½

Field at large distance - ½



½

Force on either charge F = qE

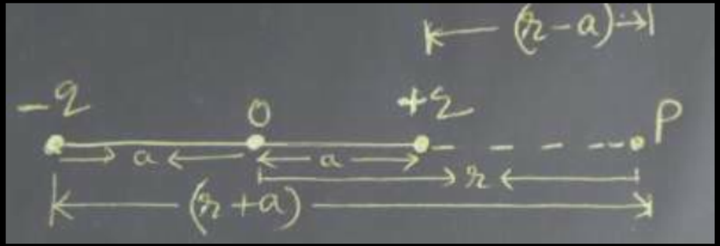
Magnitude of torque = Either of force X ⊥ distance between them.

$\tau = qE\,2a \sin\theta$ ½

$\tau = pE \sin\theta$ ½

$\vec{\tau} = \vec{p} \times \vec{E}$

Direction is normal to the paper coming out of it. ½



½

$$E_- = \frac{q}{4\pi\varepsilon_0 (r+a)^2} \, along \, (-)\,\vec{p}$$ ½

$$E_+ = \frac{q}{4\pi\varepsilon_0 (r+a)^2} \, along \, \vec{p}$$

$$\therefore Total\ field\ at\ P,\ E = E_- - E_+$$

$$= \frac{q}{4\pi\varepsilon_0}\left[\frac{1}{(r-a)^2} - \frac{1}{(r+a)^2}\right]$$ ½

$$= \frac{q}{4\pi\varepsilon_0}\frac{2pr}{(r^2-a^2)^2}$$

For

$r \rangle\rangle a$

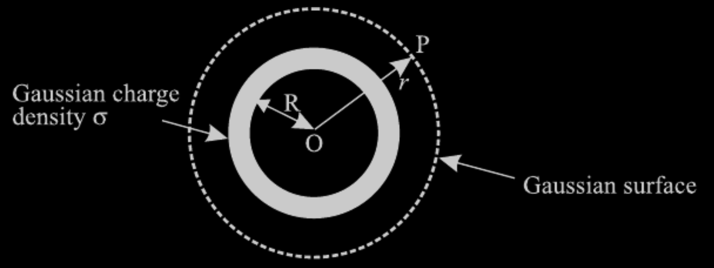
$$E = \frac{1}{4\pi\varepsilon_0}\frac{2p}{r^3}$$ ½

Apply Gauss's law to show that for a charged spherical shell, the electric field outside the shell is, as if the entire charge were concentrated at the centre.

OR

Two large parallel plane sheets have uniform charge densities $+\sigma$ and $-\sigma$. Determine the electric field (i) between the sheets, and (ii) outside the sheets.

i)	Labelled diagram	½
ii)	Flux through Gaussian surface	½
iii)	Calculation & Result	½+½



½

Flux through the small section of Gaussian surface

$$\phi = \oint \vec{E}.\vec{ds}$$

$$\therefore \phi = \oint E ds \cos\theta$$

$$\because E \parallel \vec{ds},\ \theta = 0$$

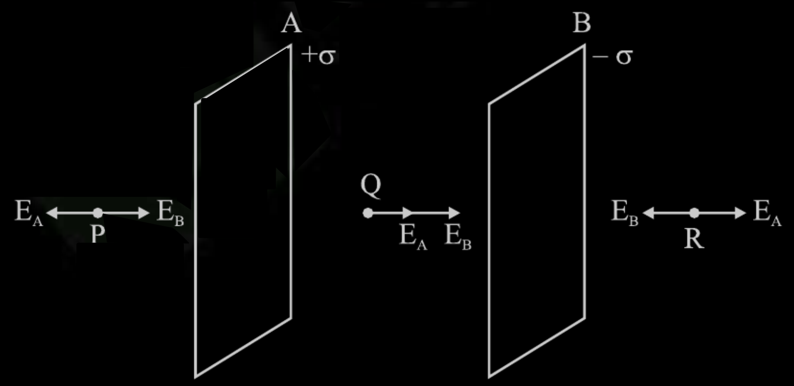
$$\therefore \phi = E.4\pi R^2 \quad \text{.......................................} (1)$$ ½

Applying Gauss's theorem

$$\phi = \frac{q}{\varepsilon_0} \quad \text{.......................................} (2)$$

from equations 1 and 2

$$E = \frac{1}{4\pi\varepsilon_0} \cdot \frac{q}{R^2}$$ ½



½

Now Electric field Intensity due to a plane sheet of charge

$$E = \frac{\sigma}{2\varepsilon_0}$$ ½

Here

$$E_A = \frac{+\sigma}{2\varepsilon_0} \text{ and } E_B = \frac{-\sigma}{2\varepsilon_0}$$

(i) Electric field at Point Q (In between the sheets)

$$\vec{E} = \vec{E}_A + \vec{E}_B = \frac{\sigma}{2\varepsilon_0} + \frac{\sigma}{2\varepsilon_0} = \frac{\sigma}{\varepsilon_0}$$ ½

(ii) Field at the point P or R

$$\vec{E} = \vec{E}_A + \vec{E}_B = \frac{\sigma}{2\varepsilon_0} - \frac{\sigma}{2\varepsilon_0} = 0$$ ½

Two identical conducting balls A and B have charges –Q and +3Q respectively. They are brought in contact with each other and then separated by a distance d apart. Find the nature of the Coulomb force between them.

OR

A metallic spherical shell has an inner radius R_1 and outer radius R_2. A charge Q is placed at the centre of the shell. What will be the surface charge density on the (i) inner surface, and (ii) outer surface of the shell ?

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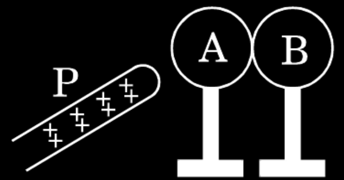
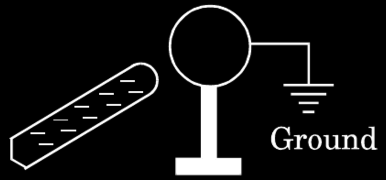
Repulsive	1
OR	
Surface charge density on inner surface $= -\dfrac{Q}{4\pi R_1^2}$	½
Surface charge density on Outer surface $= +\dfrac{Q}{4\pi R_2^2}$	½

Two metallic spheres A and B kept on insulating stands are in contact with each other. A positively charged rod P is brought near the sphere A as shown in the figure. The two spheres are separated from each other, and the rod P is removed. What will be the nature of charges on spheres A and B ?



OR

A metal sphere is kept on an insulating stand. A negatively charged rod is brought near it, then the sphere is earthed as shown. On removing the earthing, and taking the negatively charged rod away, what will be the nature of charge on the sphere ? Give reason for your answer.



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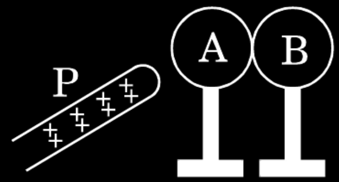

OR

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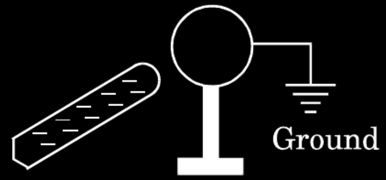
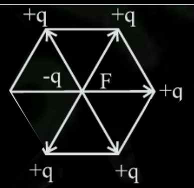


Sphere A will be negatively charged	½
Sphere B will be positively charged	½
Alternatively- B will be similarly charged to the rod and A will be oppositely charged.	½+½
OR	
Sphere will be positively charged.	½
Reason - Electrostatic Induction	½

Five point charges, each of charge $+q$ are placed on five vertices of a regular hexagon of side 'l'. Find the magnitude of the resultant force on a charge $-q$ placed at the centre of the hexagon.

OR

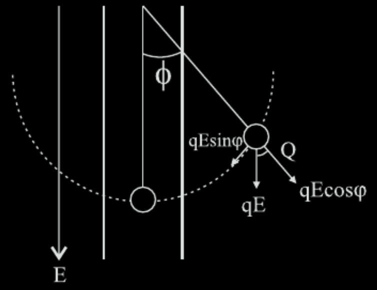
A simple pendulum consists of a small sphere of mass m suspended by a thread of length l. The sphere carries a positive charge q. The pendulum is placed in a uniform electric field of strength E directed vertically downwards. Find the period of oscillation of the pendulum due to the electrostatic force acting on the sphere, neglecting the effect of the gravitational force.



Alternatively

The forces due to the charges placed diagonally opposite at the vertices of hexagon, on the charge -q cancel in pairs. Hence net force is due to one charge only. 1

Net Force $|\bar{F}| = \dfrac{1}{4\pi\varepsilon_0}\dfrac{q^2}{l^2}$ 1



½

Derivation

$F_r = -qE\sin\varphi$ (Restoring force) ½

$ma = -qE\sin\varphi$

when φ is small

$ma = -qE\varphi$

$$m\frac{d^2x}{dt^2} = -qE\,\frac{x}{l}$$

$$\frac{d^2x}{dt^2} = -q\,\frac{E}{m}\,\frac{x}{l}$$ ½

comparing with equation of linear SHM.

$$\frac{d^2x}{dt^2} = -\omega^2 x$$

$$\omega = \sqrt{\frac{qE}{ml}}$$

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{ml}{qE}}$$ ½

Alternatively - The student can use angular SHM expression also. Full marks to be awarded for correct answer even without intermediate steps.

Two large charged plane sheets of charge densities σ and -2σ C/m^2 are arranged vertically with a separation of d between them. Deduce expressions for the electric field at points (i) to the left of the first sheet, (ii) to the right of the second sheet, and (iii) between the two sheets.

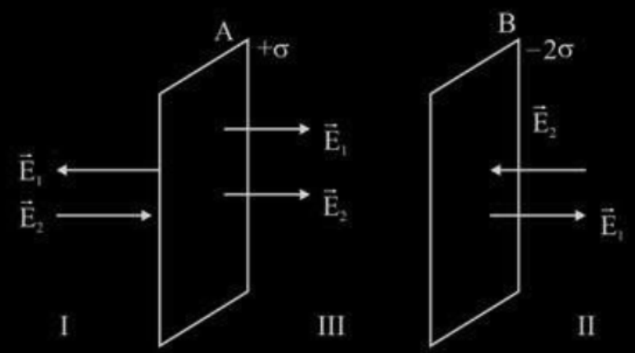
OR

A spherical conducting shell of inner radius r_1 and outer radius r_2 has a charge Q.

(a) A charge q is placed at the centre of the shell. Find out the surface charge density on the inner and outer surfaces of the shell.

(b) Is the electric field inside a cavity (with no charge) zero; independent of the fact whether the shell is spherical or not ? Explain.

Diagram



1½

Electric field in the region left of first sheet

$E_I = E_1 + E_2$

$E_I = \frac{\sigma}{\varepsilon_0} - \frac{\sigma}{2\varepsilon_0}$

$E_I = +\frac{\sigma}{2\varepsilon_0}$ ½

It is towards right

Electric field in the region to the right of second sheet

$E_{II} = \frac{\sigma}{2\varepsilon_0} - \frac{\sigma}{\varepsilon_0}$

$E_{II} = -\frac{\sigma}{2\varepsilon_0}$ ½

It is towards left

Electric field between the two sheets

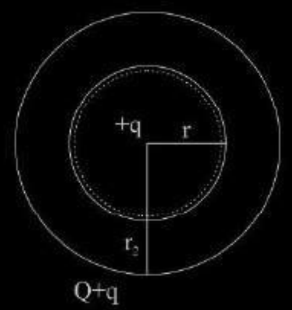
$E_{III} = E_1 + E_2$

$E_{III} = \frac{\sigma}{\varepsilon_0} + \frac{\sigma}{2\varepsilon_0}$

$E_{III} = \frac{3\sigma}{2\varepsilon_0}$ ½

Electric field is towards the right

(a) Diagram



½

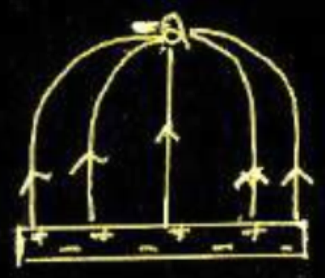
The surface charge density on inner surface of the shell is $\sigma_1 = -\frac{q}{4\pi r_1^2}$ 1

The surface charge density on outer shell is $\sigma_2 = \frac{Q+q}{4\pi r_2^2}$ ½

(b) Consider a Gaussian surface inside the shell, net flux is zero since $q_{net} = 0$. According to Gauss's law it is independent of shape and size of shell. 1

Draw the pattern of electric field lines, when a point charge – Q is kept near an uncharged conducting plate.

Draw the pattern of electric field lines, when a point charge – Q is kept near an uncharged conducting plate.



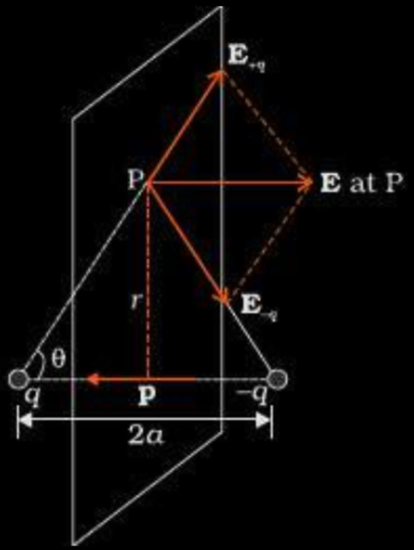
1

[Note: i) Deduct ½ mark, if arrows are not shown.
ii) do not deduct any mark, if charges on the plates are not shown]

(a) Derive an expression for the electric field at any point on the equatorial line of an electric dipole.

(b) Two identical point charges, q each, are kept 2m apart in air. A third point charge Q of unknown magnitude and sign is placed on the line joining the charges such that the system remains in equilibrium. Find the position and nature of Q.

(a)



1

The magnitude of the electric fields due to the two charges +q and -q are

$$E_{+q} = \frac{1}{4\pi \epsilon_0} \frac{q}{\left(r^2 + a^2\right)}$$ ½

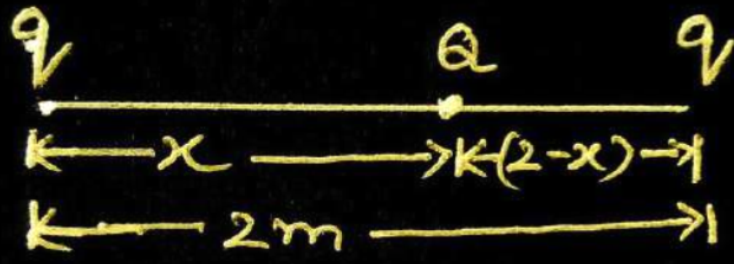
$$E_{-q} = \frac{1}{4\pi \epsilon_0} \frac{q}{\left(r^2 + a^2\right)}$$ ½

The components normal to the dipole axis cancel away and the components along the dipole axis add up

Hence total Electric field = - ($E_{+q} + E_{-q}$)cosθ $\hat{p}$ ½

$$E = -\frac{2qa}{4\pi\varepsilon_0\left(r^2 + a^2\right)^{3/2}}\ \hat{p}$$ ½

(b)



½

System is in equilibrium therefore net force on each charge of system will be zero.

For the total force on 'Q' to be zero

$$\frac{1}{4\pi \epsilon_0} \frac{qQ}{x^2} = \frac{1}{4\pi \epsilon_0} \frac{qQ}{(2-x)^2}$$ ½

x = 2 – x

½

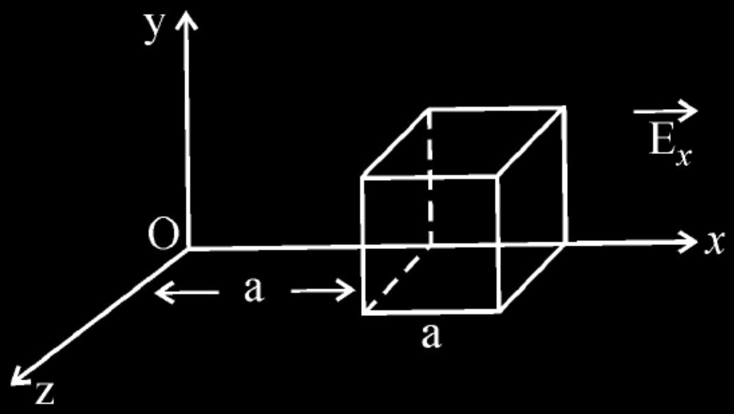
2x = 2

x = 1 m

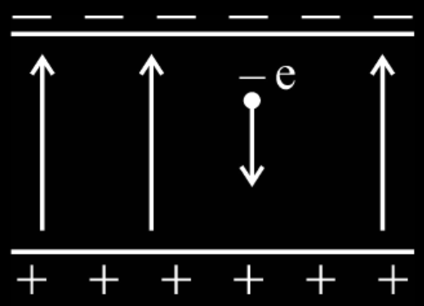
(Give full credit of this part, if a students writes directly 1m by observing the given condition)

For the equilibrium of charge "q" the nature of charge Q must be opposite to the nature of charge q. ½

Define electric flux and write its SI unit. The electric field components in the figure shown are : $E_x = \alpha x$, $E_y = 0$, $E_z = 0$ where $\alpha = \frac{100\ N}{Cm}$. Calculate the charge within the cube, assuming a = 0.1m.

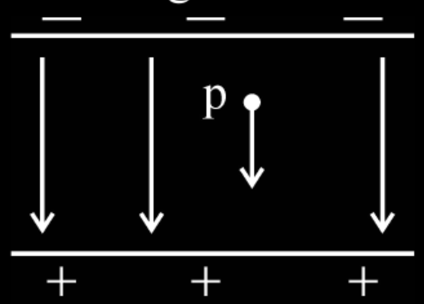


An electron falls through a distance of 1.5 cm in a uniform electric field of magnitude 2.0×10^4 N/C (Fig. a)



(a)

Calculate the time it takes to fall through this distance starting from rest.



(b)

If the direction of the field is reversed (fig. b) keeping its magnitude unchanged, calculate the time taken by a proton to fall through this distance starting from rest.

(a) Derive an expression for the electric field E due to a dipole of length '2a' at a point distant r from the centre of the dipole on the axial line.

(b) Draw a graph of E versus r for r >> a.

(c) If this dipole were kept in a uniform external electric field E_0, diagrammatically represent the position of the dipole in stable and unstable equilibrium and write the expressions for the torque acting on the dipole in both the cases.

(a)

Electric field at P due to charge $(+q) = E_1 = \frac{1}{4\pi\varepsilon_0}\frac{q}{(r-a)^2}$ ½

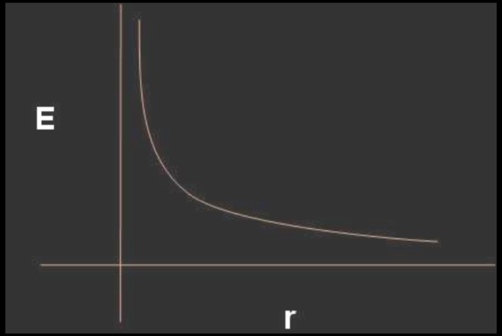
Electric field at P due to charge $(-q) = E_2 = \frac{1}{4\pi\varepsilon_0}\frac{q}{(r+a)^2}$ ½

Net electric Field at P= $E_1 - E_2 = \frac{1}{4\pi\varepsilon_0}\frac{q}{(r-a)^2} - \frac{1}{4\pi\varepsilon_0}\frac{q}{(r+a)^2}$ ½

$$= \frac{1}{4\pi\varepsilon_0}\frac{2pr}{(r^2-a^2)^2} \qquad (p = q.2a)$$

Its direction is parallel to $\vec{p}$. ½

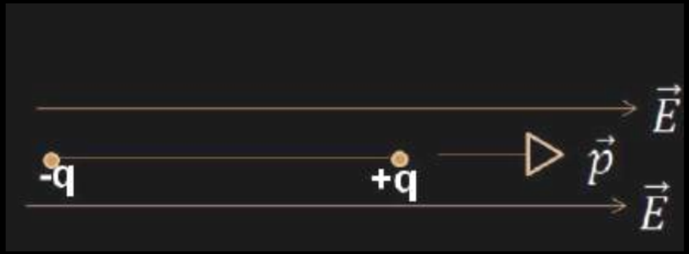
(b)



1

(Note: Award ½ mark if the student just writes: For short Dipole = $\frac{1}{4\pi\varepsilon_0}\frac{2p}{r^3}$ without drawing the graph)

(c)



Stable equilibrium ½

Unstable equilibrium ½

(Note: Award ½ mark only if the student does not draw the diagrams but just writes:

(i) For stable Equilibrium: $\vec{p}$ is parallel to $\vec{E}$.

(ii) For unstable equilibrium: $\vec{p}$ is antiparallel to $\vec{E}$)

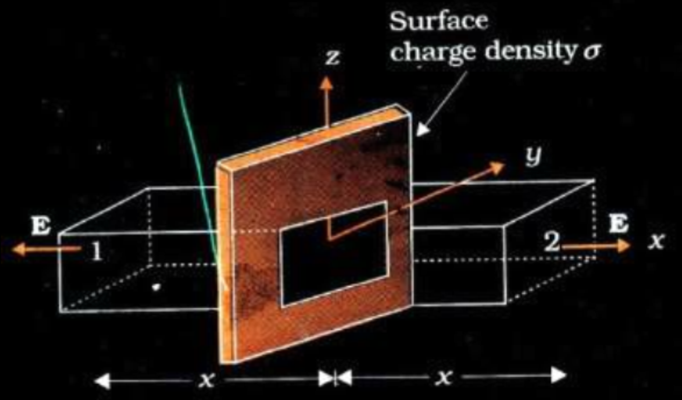
Torque = 0 for (i) as well as case (ii).

(Also accept, $\vec{\tau} = \vec{p} \times \vec{E}$ / $\tau = pE\sin\theta$) ½ + ½

(a) Use Gauss's theorem to find the electric field due to a uniformly charged infinitely large plane thin sheet with surface charge density σ.

(b) An infinitely large thin plane sheet has a uniform surface charge density $+\sigma$. Obtain the expression for the amount of work done in bringing a point charge q from infinity to a point, distant r, in front of the charged plane sheet.

a)



$$\oint E.ds = \frac{q}{\varepsilon_0}$$

½

The electric field E points outwards normal to the sheet. The field lines are parallel to the Gaussian surface except for surfaces 1 and 2. Hence the net flux $= \oint E.ds = EA + EA$ where A is the area of each of the surface 1 and 2. **1**

$$\therefore \oint E.ds = \frac{q}{\varepsilon_0} = \frac{\sigma A}{\varepsilon_0} = 2EA;$$

1

$$E = \frac{\sigma}{2\varepsilon_0}$$

½

b)

$$W = q\int_{\infty}^{r} \vec{E}.d\vec{r}$$ ½

$$= q\int_{\infty}^{r}(-Edr)$$ ½

$$= -q\int_{\infty}^{r}\left(\frac{\sigma}{2\epsilon_0}\right)dr$$ ½

$$= \frac{q\sigma}{2\in}|\infty - r|$$

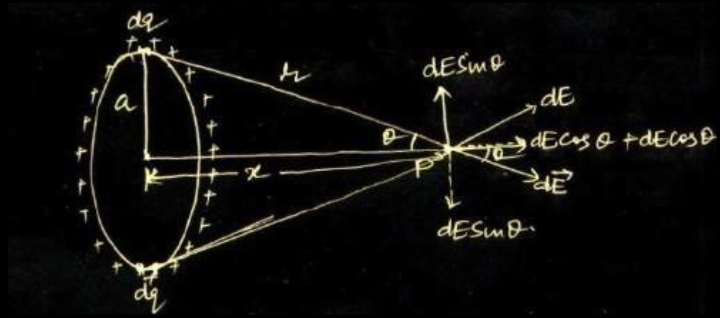
$$\Longrightarrow (\infty)$$ ½

How does the electric flux due to a point charge enclosed by a spherical Gaussian surface get affected when its radius is increased ?

How does the electric flux due to a point charge enclosed by a spherical Gaussian surface get affected when its radius is increased ?

Electric flux remains unaffected.

A charge is distributed uniformly over a ring of radius ‘a’. Obtain an expression for the electric intensity E at a point on the axis of the ring. Hence show that for points at large distances from the ring, it behaves like a point charge.



½

Net Electric Field at point P = $\int_{o}^{2\pi a} dE\cos\theta$

dE = Electric field due to a small element having charge dq ½

$= \frac{1}{4\pi\varepsilon_o}\frac{dq}{r^2}$

Let λ = Linear charge density

$= \frac{dq}{dl}$

$dq = \lambda dl$

Hence $E = \int_{o}^{2\pi a} \frac{1}{4\pi\varepsilon_o}.\frac{\lambda dl}{r^2} \times \frac{x}{r}$, where $\cos\theta = \frac{x}{r}$ ½

$= \frac{\lambda x}{4\pi\varepsilon_o r^3}(2\pi a)$

$= \frac{1}{4\pi\varepsilon_o}\frac{Qx}{(x^2+a^2)^{\frac{3}{2}}}$, where total charge $Q = \boldsymbol{\lambda}\times 2\pi a$ ½

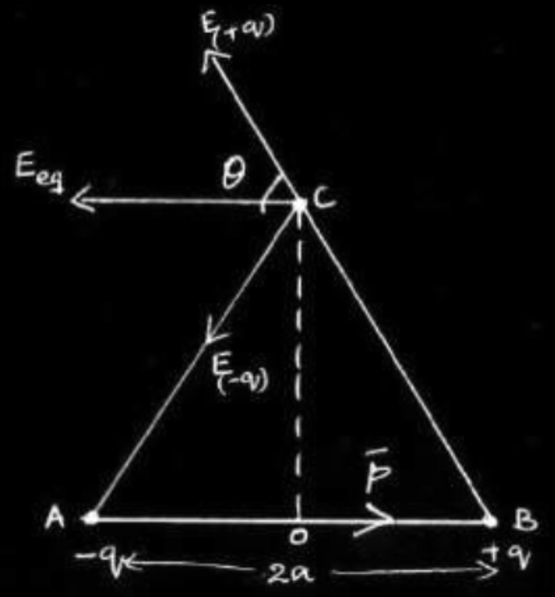
At large distance i.e. x>>a

$E \simeq \frac{1}{4\pi\varepsilon_o}.\frac{Q}{x^2}$ ½

This is the Electric field due to a point charge at distance x. ½

(**NOTE**: Award two marks for this question, if a student attempts this question but does not give the complete answer)

Derive an expression for the electric field intensity at a point on the equatorial line of an electric dipole of dipole moment $\vec{p}$ and length 2a. What is the direction of this field ?



1

$E_{+q} = \quad Kq / (r^2 + a^2)$ and $E_{-q} = \ Kq / (r^2 + a^2)$ ½

The two Electric fields have equal magnitudes and their directions are as shown in diagram

Components along dipole axis get added up while normal components cancel each other. ½

$$\therefore \mathbf{E} = -[E_{-q} + E_{+q}] \cos\theta \ \hat{r} \text{ so } E = -\frac{K2qa}{[r^2+a^2]^{\frac{3}{2}}} \hat{r}$$ ½

$$= \frac{kp}{[r^2+a^2]^{\frac{3}{2}}} (p = 2qa\hat{r}) = \frac{-1}{4\pi\epsilon_o} \frac{p}{[r^2+a^2]^{\frac{3}{2}}}$$

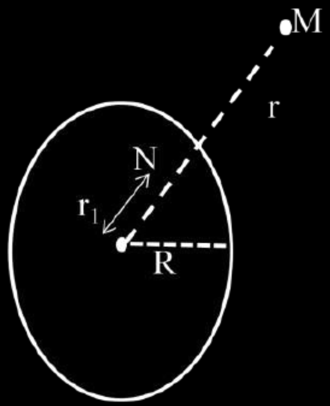
∴Direction of electric field is opposite to that of dipole moment. ½

Find the electric field intensity due to a uniformly charged spherical shell at a point (i) outside the shell and (ii) inside the shell. Plot the graph of electric field with distance from the centre of the shell.

We have by Gauss's law $\oint \vec{E}.\overrightarrow{dS} = \frac{Q_{enclosed}}{\epsilon_0}$ ½

Let Q be the total charge on the shell

(i) For the point M outside the shell, we have



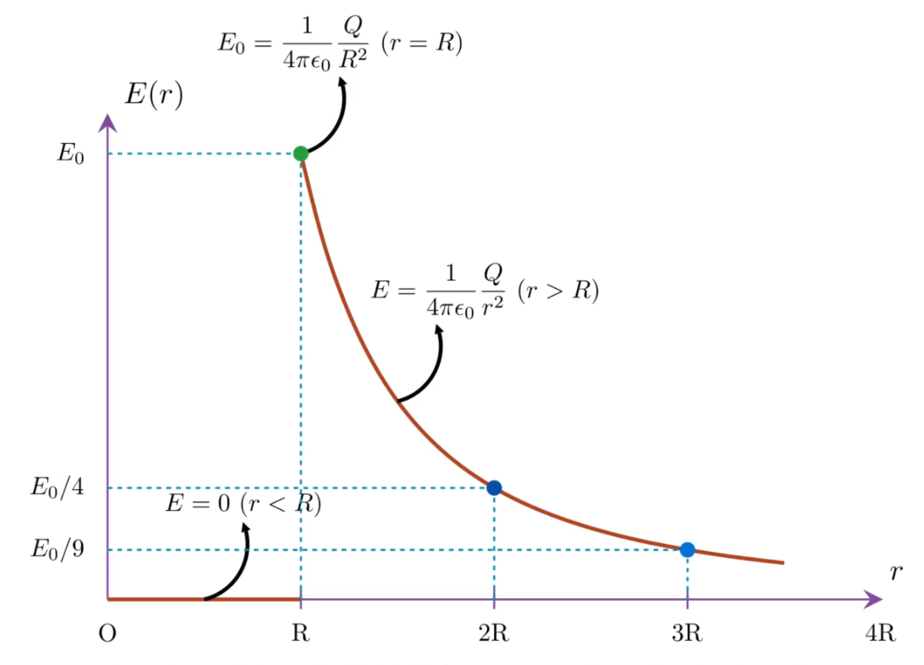
$$E.4\pi r^2 = \frac{Q}{\epsilon_0}$$ ½

$$\therefore E = \frac{1}{4\pi\epsilon_o}\frac{Q}{r^2}$$ ½

(ii) For the point N inside the shell, as charge enclosed inside the shell is zero. ½

$$E.4\pi r_1^2 = 0$$ ½

$$\therefore E = 0$$



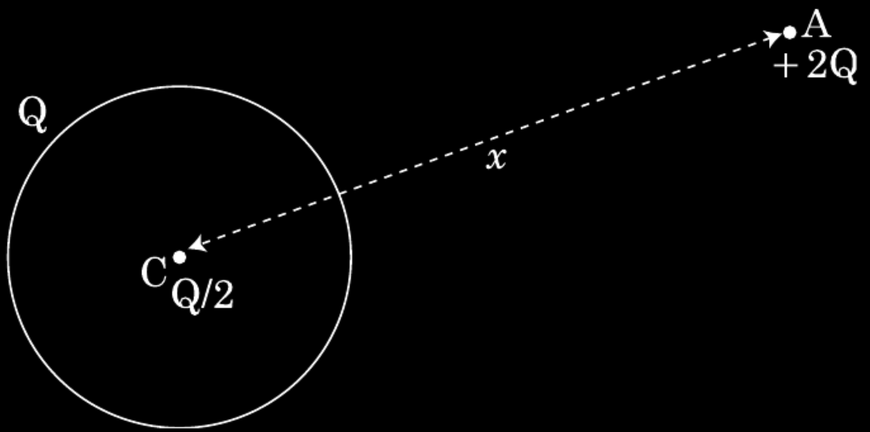
Graph of Electric field (E) versus r for spherical shell.

A thin metallic spherical shell of radius R carries a charge Q on its surface. A point charge Q/2 is placed at the centre C and another charge $+2Q$ is placed outside the shell at A at a distance x from the centre as shown in the figure.

(i) Find the electric flux through the shell.

(ii) State the law used.

(iii) Find the force on the charges at the centre C of the shell and at the point A.



(i) Electric flux through a Gaussian surface,

$\varphi = \frac{total\ \ enclosed\ \ charge}{\epsilon_0}$ ½

Net charge enclosed inside the shell q=0

∴ Electric flux through the shell $\frac{q}{\epsilon_o}$=0 ½

Award ½ mark even when the student writes - Electric flux through the shell is zero as electric field inside the shell is zero.

(ii) Gauss Law- Electric flux through a Gaussian surface is $^1/_{\epsilon_0}$ times the net charge enclosed with in it. 1

Alternatively, $\oint \vec{E}.\overrightarrow{dS} = \frac{q}{E_0}$

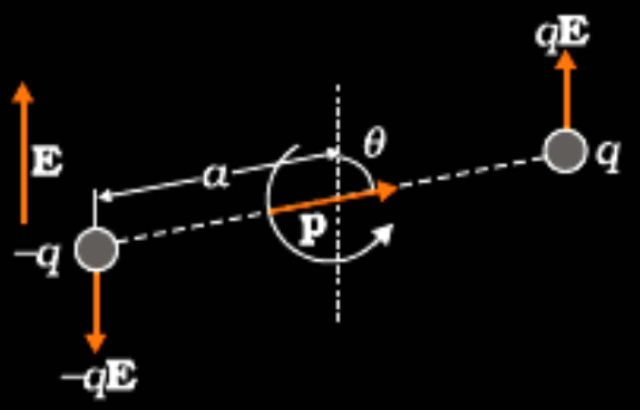
(iii) Force on the charge at the centre i.e. Charget $^Q/_2 = 0$ ½

$F_A = \frac{1}{4\pi E_0}\frac{2Q \times \left(Q + {}^Q/_2\right)}{x^2}$

$= \frac{1}{4\pi E_0}\frac{3Q^2}{x^2}$ ½

(a) Deduce the expression for the torque acting on a dipole of dipole moment $\vec{p}$ placed in a uniform electricfield $\vec{E}$. Depict the direction of the torque. Express it in the vector form.

(a)



	½
Magnitude of torque = magnitude of either force multiplied by the arm of the couple. $= qE \times 2a\sin\theta$	½
$= pE\sin\theta$	½
Direction of torque is perpendicular to the plane containing $\vec{p}$ *and* $\vec{E}$.	½
Vector form $\vec{\tau} = \vec{p} \times \vec{E}$	½